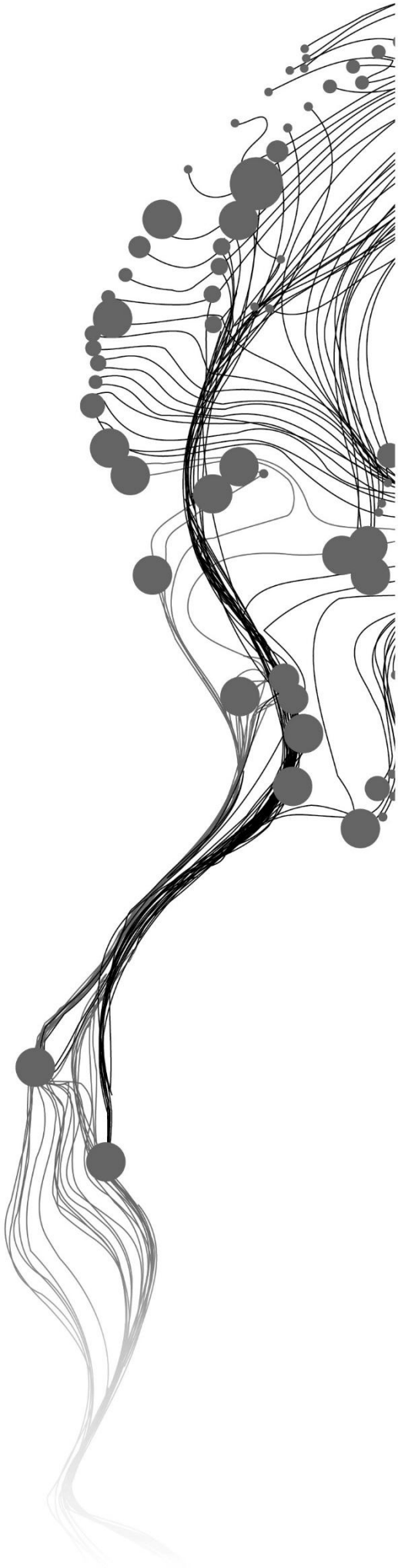
SPATIO-TEMPORAL MODELLING OF CLIMATE-DRIVEN CHOLERA RISK IN A DATA UNCERTAINTY ENVIRONMENT

COURDET KUKO AZONG
July, 2026

SUPERVISORS:
Prof. dr. Marc van den Homberg
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DISCLAIMER

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ABSTRACT

Cholera remains a persistent public health burden in Cameroon, where approximately 18,986 cases and 478 deaths were reported between 2018 and 2023. Despite this burden, the spatial and temporal dynamics of cholera transmission at the health district level remain insufficiently understood, and the application of Bayesian spatio-temporal modelling approaches in Cameroon remains limited. This study investigated the spatio-temporal distribution of cholera risk across approximately 200 health districts in Cameroon between 2018 and 2023 using Bayesian disease mapping models implemented through Integrated Nested Laplace Approximation (INLA).

The study adopted a two-stage modelling frameworks. First, autoregressive temporal models were used to investigate temporal dependence and the contribution of hydroclimatic variables to cholera dynamics at the regional level. Second, the Bayesian spatio-temporal disease modelling approach, incorporating spatial random effects, temporal processes, environmental, hydroclimatic, socioeconomic, and healthcare-related covariates were developed at the health district level. Model performance was assessed using the Deviance Information Criterion (DIC) and predictive accuracy metrics.

The results revealed strong temporal dependence in cholera incidence, indicating that transmission in a given week was strongly influenced by incidence in immediate past weeks. The inclusion of rainfall and temperature explained part of the temporal variation in several regions, although some unexplained variability remained. Among the district-level models, the full model incorporating spatial and temporal random effects alongside covariates achieved the best fit and hence was used to make inferences. Decreased cholera incidence was associated with higher slopes (β: -0.112; 95%CrI: -0.201, -0.024), higher poverty (β: -1.459; 95%CrI: -2.417, -0.489), and unimproved sanitation (β: -0.059; 95%CrI: -3.154, 3.028). Whereas, increased incidence was associated with higher water presence (β: 2.822; 95%CrI: 0.526, 5.112) and healthcare facilities per 1000 population (β: 0.884; 95%CrI: -1.180, 2.963). While rainfall and temperature exhibited nonlinear relationships that varied across regions. Hence the conclusion was that including these covariates improved model fitness to the data, and explained some spatial variation of the risk spread. Furthermore, persistent hotspots were identified in the southern coastal regions and northern border districts, with exceedance probability maps highlighting areas of consistently elevated risk throughout the study period.

These findings demonstrate that cholera transmission in Cameroon is shaped by temporal, spatial, environmental, and contextual factors. The study highlights the value of Bayesian spatio-temporal disease mapping for identifying priority intervention area and thereby supporting anticipatory action and preparedness planning.

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LIST OF ABBREVIATIONS

ACF	Autocorrelation Function
PACF	Partial Autocorrelation Function
ACLED	Armed Conflict Location and Event Data
AR	Autoregressive
BYM	Besag York Mollié
DIC	Deviance Information Criterion
GTFCC	Global Task Force on Cholera Control
iCAR	Intrinsic Conditional Autoregressive
iid	Independent and Identically Distributed
INLA	Integrated Nested Laplace Approximations
MCMC	Markov Chain Monte Carlo
RW	Random Walk
SMR	Standardised Morbidity Ratio
VBNC	Viable but Non-culturable
WASH	Water, Sanitation, and Hygiene
WHO	World Health Organisation
VIF	Variance Inflation Factor

1. INTRODUCTION

1.1. Background

Despite decades of public health intervention efforts, cholera remains one of the most persistent waterborne infectious diseases globally. The disease is caused by the bacterium Vibrio cholerae and is primarily transmitted through the ingestion of contaminated water or food (Ganesan et al., 2020). Cholera results in symptoms such as acute watery diarrhoea, which can rapidly progress to severe dehydration and death if left untreated (Montero et al., 2023). Although both preventable and treatable, cholera continues to disproportionately affect vulnerable populations in the global South, who lack access to safe water, sanitation, and healthcare (Ganesan et al., 2020).

Low- and middle- income countries face the much of the cholera burden. Recent outbreaks have had an increasing trend in geographical spread and intensity (Mohamed et al., 2025). Additionally, a rise in mortality rates associated with climatic extremes, humanitarian crises, rapid urbanisation, and fragile health systems (Dereje et al., 2025). According to the World Health Organisation (WHO), Africa accounted for a major proportion of the global cholera cases and deaths, recording approximately 46 percent of global cholera cases and nearly 83 percent of deaths in 2024 (WHO, 2025).

Cameroon has not been exempted from this burden. Since the first documented cholera outbreak in the country in 1971, recurrent epidemics have occurred across multiple decades, affecting both urban and rural areas (Ngwa et al., 2016). Between 2000 and 2012, 43,474 cases were reported, with a mean annual incidence rate of 17.78 per 100,000 population and a case fatality ratio of 4 percent, far exceeding the WHO threshold of less than 1%. More recently, between 2018 and 2023, approximately 18,986 cases and 478 deaths were recorded, with over 80 percent of cases concentrated in the Littoral, South-West, and South regions (Mboringong et al., 2025). Historical outbreaks have shown that cholera transmission in the country is often concentrated within the coastal, flood-prone, densely populated regions, areas affected by conflicts, and areas that practice open defecation (Mboringong et al., 2025; Musa et al., 2022).

The dynamics of cholera transmission, is driven by several environmental and socio-economic drivers and their interactions within space and time (Usmani et al., 2021). These drivers provide favourable conditions for the bacteria to grow and spread. Environmental factors such as prolonged high temperatures provide favourable conditions for the bacteria to grow, while other factors such as extreme rainfall, flooding, and drought enable the spread of the disease in the environment through water surface contamination (Usmani et al., 2021). These factors disproportionately affect communities characterised by high population and urban density, unimproved water and sanitation infrastructure, and poor access to healthcare services (Mohamed et al., 2025). Such conditions are present in Cameroon, where inadequate WASH infrastructure in rapidly growing informal settlements has been exacerbated by the displacement of populations due to ongoing conflicts (Mboringong et al., 2025; Ngomba et al., 2026). At the same time, Cameroon's fragile healthcare and surveillance systems struggle to cope with large outbreaks due to limited healthcare capacity, delayed reporting, and inadequate resources for testing (Ateudjieu et al., 2019; Yakum et al., 2017).

Cholera prevention and control strategies are typically implemented across three broad and often overlapping phases: seasonal preparedness, anticipatory action, and outbreak response. Seasonal

preparedness focuses on strengthening surveillance systems, pre-positioning medical supplies, improving water, sanitation and hygiene (WASH) infrastructure, and identifying populations at elevated risk before transmission intensifies (Alcayna et al., 2025). Anticipatory action aims to initiate interventions before outbreaks occur, often by predefined triggers derived from surveillance indicators, environmental conditions, or climatic forecasts.

Modelling approaches play different roles across these intervention phases. Risk and vulnerability models, such as those developed within the Priority Areas for Multisectoral Interventions (PAMI) framework and the INFORM Epidemic Risk Index, are commonly used during preparedness planning to identify geographic areas with elevated epidemic risk (Kiama et al., 2025; Poljanšek et al., 2018). In contrast, trigger-based models that are increasingly used to support anticipatory action by identifying conditions that may lead to outbreaks. For example, the Cameroon Cholera Early Action Protocol incorporates predefined triggers linked to surveillance and context specific indicators to guide the timely implementation of preventive measures (Red Cross Red Crescent Movement, 2025; Van Den Homberg et al., 2025). During outbreak response, epidemiological models can support resource allocation and intervention targeting, while explanatory models can improve understanding of the environmental and socio-economic drivers that influence disease transmission. Such information is critical for identifying the factors associated with cholera risk and informing evidence-based intervention strategies.

Since John Snow's mapping of cholera cases in 1854, a wide range of modelling approaches have been applied to infectious disease analysis and prediction. These include mechanistic compartmental models, agent-based models, machine learning approaches, and statistical disease mapping frameworks. (Campbell et al., 2020; Stevens & Pfeiffer, 2011; Wang, 2022). Such models have improved understanding of how diseases spread and how environmental and social factors influence transmission. For example, Kapwata & Gebreslasie, (2016) applied a random forest spatial modelling approach to malaria transmission in South Africa and found that environmental factors such as altitude, NDVI, temperature, and lagged rainfall were among the most influential predictors in explaining spatial patterns of disease risk. Another example is the application of agent-based modelling on understanding COVID-19 transmission by Shamil et al. (2024), which simulated interactions between individuals within a city to analyse disease transmission dynamics and evaluate the effectiveness of interventions such as lockdowns and contact tracing.

Added to the above, Bayesian hierarchical models have increasingly been applied in spatio-temporal disease modelling to represent how model parameters affect model outcomes along with their uncertainties (Anteneh et al., 2024). An additional strength of Bayesian modelling is the ability to incorporate priors based on existing epidemiological or expert knowledge of model parameters to improve prediction (Cook et al., 2023). This is particularly important in settings where there are inconsistencies in the quality of surveillance data. Several studies have demonstrated the strengths of Bayesian spatio-temporal approaches in infectious disease modelling. For example, Njotto et al. (2025) applied spatial and temporal dependency (Besag York Mollié and random walk respectively) priors to model malaria incidence risk in Tanzania and identified significant environmental drivers associated with disease risk. Similarly, Cook et al. (2023) demonstrated that the use of prior knowledge from a study conducted in a neighbouring state on wasting disease led to valid predictions of the disease in a limited local data setting. However, the Bayesian model application to cholera transmission in Cameroon remains relatively underexplored.

Therefore, this study aims to develop a Bayesian Spatio-temporal model to predict cholera risk by accounting for the covariates and spatial and temporal dependencies in cholera transmission across Cameroon. By identifying the factors associated with cholera risk and quantifying their effects through

time and space, the study seeks to generate evidence that can support preparedness planning, anticipatory action, and targeted intervention strategies. To achieve this aim, the study leverages a unique multi-source dataset comprising eight years of weekly cholera surveillance records (2017-2024), high-resolution satellite-derived climatic and flood indicators, and district-level structural vulnerability metrics. Access to these complementary datasets provides a novel level of temporal and spatial detail for Cameroon, thereby strengthening the feasibility of the research and enabling analyses that have not previously been undertaken in this setting.

1.2. Problem statement

Cholera outbreaks in low- and middle-income countries often follow two epidemiological patterns: endemic or epidemic (Usmani et al., 2021). In endemic settings, such as Bangladesh, cholera persists year-round with predictable seasonal peaks (Codeço, 2001). Bangladesh displays consistent seasonal cycles that support strong forecasting capability (Brumfield et al., 2025; Sack et al., 2021). In contrast, epidemic settings experience sporadic outbreaks after long periods of apparent absence, often triggered by climatic shocks, population displacement, or unimproved water and sanitation systems (Codeço, 2001). However, the extent to which these factors contribute to cholera transmission dynamics across Cameroon remains insufficiently understood, particularly at the health district spatial scale.

A major challenge in cholera preparedness and response is the uncertainty associated with surveillance data, including underreporting and reporting delays(Olu et al., 2025). Consequently, many cholera early warning and anticipatory action systems rely on trigger based approaches that use environmental and hydroclimatic indicators associated with cholera emergence, persistence and spread (Usmani et al., 2021). While these approaches provide a practical framework for early action, they primarily focus on predefined environmental thresholds and may not fully capture the broader spatial and temporal transmission dynamics of cholera or the complex interactions between environmental and socio-economic drivers.

Furthermore, the application of Bayesian spatial and temporal disease mapping approaches to cholera modelling in Cameroon remains limited. Most existing studies have largely relied on descriptive epidemiology and conventional statistical methods, with limited consideration of the spatial and temporal dependence that characterises cholera transmission (Bangwen et al., 2024; Mboringong et al., 2024, 2025; Musa et al., 2022). Neighbouring districts may experience similar cholera risks due to shared environmental conditions, sanitation infrastructure, climatic variability, transportation networks and population mobility patterns. Failure to account for these dependencies may lead to an incomplete understanding of cholera risk patterns and limit the effectiveness of targeted interventions.

Bayesian modelling frameworks offer additional advantages beyond accounting for spatial and temporal dependence. They provide a principled approach for quantifying uncertainty and allow the incorporation of prior epidemiological knowledge, which may improve model stability and predictive performance, particularly in settings where surveillance data are sparse or incomplete (Anteneh et al., 2024; Cook et al., 2023). However, limited evidence exists regarding the contribution of temporal processes and environmental covariates in explaining spatial and temporal variations in cholera risk in Cameroon.

Consequently, an important research gap exists in understanding the contribution of explanatory variables, spatial dependence structures, and temporal dynamics in explaining cholera transmission across Cameroon. Addressing this gap is important for improving understanding of cholera epidemiology and strengthening anticipatory outbreak preparedness systems. Using openly accessible datasets from sources including Climate Hazards Group InfraRed Precipitation with Station Data (CHIRPS), Fifth Generation European Centre for Medium-Range Weather Forecasts (ECMWF) Atmospheric Reanalysis (ERA5), WorldPop, and the Humanitarian Data Exchange, the study assesses how these datasets contribute to observed cholera transmission patterns across health districts.

1.3. Research objectives

To investigate the spatial and temporal dynamics of cholera risk in Cameroon using Bayesian spatio-temporal disease mapping models and to assess the impact of model parameters in explaining variations in cholera risk across space and time.

1.3.1. Sub-objectives

1. To evaluate the temporal autoregressive process governing localised cholera transmission.
2. To assess the effects of selected hydroclimatic, environmental, and socio-economic covariates on the spatio-temporal distribution of cholera incidences.
3. To identify the spatial distribution of cholera residual relative risk and persistent hotspots associated with the spatially structured random effect across health districts in Cameroon.

2. LITERATURE REVIEW

This chapter includes the literature relevant to vibrio cholerae transmission, focusing on the epidemiology, hydroclimatic, environmental, and socio-economic drivers to cholera outbreak and focusing on Cameroon. As well as examining the application of Bayesian spatio-temporal models in risk prediction.

2.1. Cholera epidemiology and ecology

Cholera is an acute diarrhoeal disease caused by Vibrio cholerae O1 and O139 (Montero et al., 2023). It is transmitted through the faecal-oral route by ingestion of contaminated water or food (Montero et al., 2023). Under ideal environmental conditions, characterised by abundant nutrients, adequate oxygen, and optimal temperature and salinity, bacteria grow exponentially during the log phase until these resources become depleted (Wagley, 2023). Hence reaching the stationary phase and can lead to some dying and some entering the viable-but-non-culturable (VBNC) state where they can survive in unfavourable environmental conditions. The pathogen's environmental persistence and ability to shift between culturable and VBNC states make it particularly suited to fluctuating aquatic environment conditions (Montero et al., 2023; Wagley, 2023).

2.2. Environmental, hydroclimatic, and socio-economic drivers of cholera

Understanding the interaction of *V.cholerae* with both the environment and the human host is crucial for explaining its persistence, transmission dynamics, and outbreak potential. Environmental and hydroclimatic factors influence the survival and proliferation of the bacterium, whereas socio-economic factors shape population vulnerability and exposure, thereby influencing the risk of cholera transmission and outbreaks.

2.2.1. Hydroclimatic factors

Seasonal fluctuations in aquatic ecosystems, such as plankton blooms triggered by rising temperatures or available nutrients, can increase *V.cholerae* concentrations and heighten infection risks, particularly in communities relying on untreated water sources (Anas et al., 2021). Temperatures at around 37℃ provide favourable conditions for *V.cholerae* to grow optimally and can persist in environment between 10℃ and 43℃ temperatures (Khan et al., 2019). Elevated temperatures, especially when followed by rainfall in areas with poor WASH infrastructure, significantly increase cholera risk (Jutla et al., 2015). However, the effects of temperature on cholera incidence are often not immediate and may occur after a prolonged favourable condition, commonly referred to as a time lag (Asadgol et al., 2020). Several studies have investigated the relationship between temperature and cholera incidence at different lag periods. For example, Jutla et al. (2013) found that elevated temperatures occurring 1-2 months before an outbreak were associated with increased cholera incidence in epidemic regions of Asia and Latin America. Similar lagged effects of temperature on cholera risk have been reported in Sub-Saharan Africa, although the duration of the lag varies between settings. Increased cholera risk was observed at a five to six month lag in Mozambique (Armando et al., 2024), a four month lag in Zanzibar (Reyburn et al., 2011), and a two week lag in Cameroon (Ngwa et al., 2016).

Hydrological extremes such as flooding, heavy rainfall, and contamination of groundwater further promote the spread of *V.cholerae* into domestic water supplies (Focus Adriano et al., 2023). These extremes promote the transmission of the bacteria by contaminating water sources through faecal transmission (Perez-Saez et al., 2022). Armando et al. (2024) determined that there was a delayed rainfall effect on cholera in Mozambique, with a one-month lag of 223.3 mm of total rainfall led to a 57% increase

in risk of cholera. Similarly, an increase in rainfall by 100 mm at lag 0 months, was associated with an increase in cholera incidence by four percent in Dhaka (Daisy et al., 2020). Furthermore, prolonged droughts can intensify dependence on scarce and often unsafe water sources, amplifying transmission risk (Magers et al., 2025). The normalised difference vegetation index (NDVI) was also found to influence cholera risk with lower NDVI associated with higher risk (Armando et al., 2024). Lower NDVI often indicate lack of vegetation representing drought conditions and hence limited water sources (Armando et al., 2024).

Other environmental factors, such as natural disasters, including earthquakes and cyclones frequently disrupt water, sanitation, and hygiene (WASH) infrastructure, increasing exposure to contaminated water (Sekine & Roskosky, 2018). For instance, the 2010 Haiti earthquake triggered one of the most devastating cholera epidemics, while recurrent flooding in Mozambique and Malawi has repeatedly fuelled outbreaks (Khan et al., 2017; Nyimba, 2025). Furthermore, proportion and proximity to waterbodies contaminated from runoff from refuse dumps were also found to be associated with cholera incidence (Osei et al., 2010).

2.2.2. Socio-economic factors

Socio-economic conditions play a critical role in shaping cholera transmission by influencing both exposure to contaminated water and vulnerability to infection. High population and building densities can increase transmission risk by facilitating close contact between individuals and placing pressure on already limited water and sanitation infrastructure (Gopo et al., 2026; Penrose et al., 2010). Among the most consistent determinants of cholera are deficiencies in water, sanitation, and hygiene (WASH) infrastructure (Manaseh et al., 2023). Poor access to safe drinking water, inadequate sanitation facilities, and unhygienic practices increase population exposure to *V.cholerae*, particularly when combined with extreme hydroclimatic conditions that contaminate water sources (Adesina et al., 2026; Poljanšek et al., 2018). Using a zero-inflated negative binomial model, Mageto et al. (2025) found that households relying on surface water sources and unimproved sanitation facilities had a significantly higher risk of cholera than those with improved water and sanitation access.

Access to healthcare services is also an important determinant of cholera outcomes and control. Delayed diagnosis and treatment can increase disease severity, mortality risk, and the duration of bacterial shedding, thereby prolonging transmission within affected communities (Djouma et al., 2016). Healthcare accessibility additionally influences case detection and reporting, which are essential for effective surveillance and outbreak response (Ingelbeen et al., 2025). Demographic characteristics may further influence cholera susceptibility. Several studies have reported a higher incidence of cholera among males than females (Mboringong et al., 2024; Nanzaluka et al., 2020). Conversely, (Al-Mohanadi et al., 2022) reported a higher susceptibility among females than males in Sana'a City, Yemen, suggesting that demographic risk factors may be context-specific.

Poverty further amplifies vulnerability to cholera by limiting access to safe water, adequate sanitation, and healthcare services. Individuals living in low-income communities and informal settlements are often disproportionately exposed to contaminated water sources and poor hygienic conditions (Charnley et al., 2021; Oweibia et al., 2025). Similarly, conflict and population displacement can accelerate disease transmission by creating overcrowded living conditions with inadequate WASH infrastructure and limited access to healthcare services (Musa et al., 2022). Together, these findings demonstrate that socio-economic factors influence not only exposure to *V.cholerae* but also the capacity of communities to prevent, detect, and respond to outbreaks.

2.3. Cholera burden in Cameroon

Cameroon is a low- and middle income country which faces sporadic cholera outbreaks across years. Since the 1971 first cholera occurrence in Cameroon, the country has experienced outbreaks which are sporadic on many occasions. The country experienced one of the most challenging outbreaks in 2010, which led to 10,759 confirmed cases and 657 deaths (Djomassi et al., 2013). Since then, the cases have occurred within years with zero cases in some years or some cases within some years. This pattern changed after 2018, when a series of epidemic peaks were recorded between 2018 and 2023, infecting 18,986 persons and leading to 478 deaths (Mboringong et al., 2025). These outbreaks challenged the health systems in Cameroon due to the unpreparedness of the health care facilities in handling outbreaks (Ateudjieu et al., 2019).

The outbreaks in Cameroon have consistently affected several regions including Littoral, Centre, Far North, and South West regions (Mboringong et al., 2024; Ngwa et al., 2016). Several factors have been identified to be potential drivers of risk across these regions. For start, rainfall is one of the main drivers of cholera in Cameroon, particularly in the in the rainy season as seen in Figure 1 (Mboringong et al., 2025; Ngwa et al., 2016). This is particularly seen in the South West and Littoral regions as it leads to flooding in some areas, while drier conditions in the Northern regions increase cholera risk (Mboringong et al., 2024).

Furthermore, Cameroon is comprised of four sub-climatic zones characterised (Figure 1). The Sudano-Sahelian and Tropical Humid climates are characterised by dry warm weather conditions with occasional flooding resulting in high cases during the rainy season (July to September) in districts in the Far North and North regions found in this zone (Djomassi et al., 2013; Ngwa et al., 2016). While the Guinea Equatorial and Equatorial Monsoon are characterised by wet conditions and humid conditions with some districts bordering the ocean and hence flatter areas prone to flooding, and experiencing the lowest cases during rainy season (Djomassi et al., 2013; Ngwa et al., 2016). The country also experiences warm temperatures which have also been shown to have a positive influence in cholera transmission in Cameroon, with Ngwa et al. (2016) determining that a national increase in temperature by 1 ℃, would result in a 42% increase in cholera risk.

Furthermore, several permanent waterbodies exist in Cameroon, which serve as habitats for the bacteria (Djaouda et al., 2020). Some of these include the lake Chad, which crosses through Cameroon and Chad (Djomassi et al., 2013). These waterbodies serve as water sources for populations with poor access to water, hence exposing the population to cholera risk (Nkum et al., 2023). Additionally, the country faces challenges in WASH access and water treatment, and some districts, such as Bibémi in the northern region, practice open defecation due to poverty prevalence and lack of access to WASH infrastructure (Djaouda et al., 2020; Ngomba et al., 2026). These conditions are worsened due to the conflicts in Cameroon in the Far North region (Boko haram) and North West and South West regions (Anglophone crisis) which led to the internal displacement of individuals in the country (Mboringong et al., 2025). This displacement led to the overcrowding of individuals in urban cities as in the case in the South West region where displaced people moved to the urban districts of Limbe, Tiko, and Buea, leading to overcrowding and limited water and sanitation facilities for these individuals (Bangwen et al., 2024; Musa et al., 2022). This rise in urbanisation is characterised by poor sanitation practices and contaminated water sources (Zerbo et al., 2020).

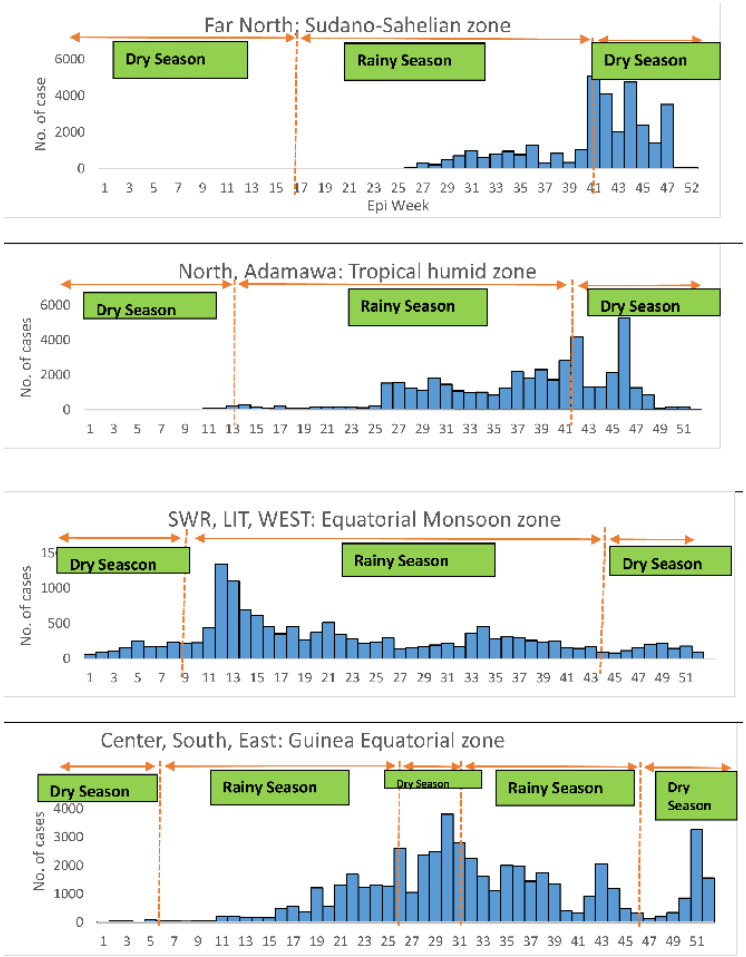


Figure 1: Number of cholera cases by season and sub-climatic zone in Cameroon, during 2018-2023. This image was derived from Mboringong et al. (2024)

Assessing risk in Cameroon is complex due to its inconsistent and sporadic cholera outbreaks. Several assessments have been carried out in Cameroon. In line with the Global Task Force for Cholera Control (GTFCC) goal of eliminating cholera by 2030, Cameroon identified priority areas in need of targeted interventions. In this process, 48 districts were identified as priority areas for multisectoral interventions (PAMI) based on cholera surveillance and vulnerability factors defined by the GTFCC (Ngomba et al., 2026). In a different approach, the European Union and JRC suggested the calculation of disease risk using several risk factors falling under three dimensions: Hazard and exposure, vulnerability, and lack of coping capacity (Poljanšek et al., 2018). This was also applied by Ifejube et al. (2023) who assessed cholera risk across Cameroon's administrative regions.

Subsequently, the Cameroon Red Cross, in collaboration with the French Red Cross and the Netherlands Red Cross' data and digital team 510, developed an early action cholera trigger model for Cameroon, incorporating thresholds such as flooding events affecting more than 2,000 people and periods of heavy rainfall defined as four consecutive days with an average daily rainfall of at least 50 mm as trigger one. This is followed by a second trigger threshold of epidemiological alerts in a district after it has experienced trigger one. Trigger two includes reporting of at least 5 cases suspected or confirmed cholera cases in the same week or at least one confirmed case in neighbouring districts (Red Cross Red Crescent Movement, 2025; Van Den Homberg et al., 2025). These trigger models can be improved by knowledge on effects of drivers on cholera transmission.

Cholera in Cameroon follows a spatial trend with disease occurring in several districts clusters, indicating spatial corelation between these areas. These areas may share cholera driving characteristics and movement of people within these areas leading to the spread. The temporal trend and persistence of the disease also suggest that the bacteria spread in the country is influenced by cases from the previous time periods. All these factors motivate for the need to understand the association of these factors to cholera incidence in Cameroon.

2.4. Space-time modelling

2.4.1. Temporal modelling

Temporal modelling approaches have been widely applied in epidemiological studies to investigate the trends in infectious disease transmission over time. Time series models are particularly important for diseases such as cholera, where transmission dynamics are influenced by previous outbreak conditions. These include; environmental persistence, variations in climatic conditions, and delayed human exposure pathways (Reyburn et al., 2011).

Autoregressive (AR) models are among the most commonly used approaches for modelling temporal dependence in epidemiological time series. These models assume that observations at a given time point are linearly dependent on previous observations, together with a random error component (Garza-Ulloa, 2018). There are different autoregressive order, that is the order in which the time periods correlate with each other (GeeksforGeeks, 2023). The first-order autoregressive process, AR(1), is particularly common because it assumes that disease incidence at time, t, depends directly on incidence observed at time $t-1$. Several Bayesian epidemiological studies have used AR(1) structures to model weekly or monthly disease persistence and outbreak evolution. For example, Charnley et al. (2021) the AR(1) model to model the relationship between drought and cholera in Africa, and found a significant association between them.

In addition to autoregressive processes, random walk models are also frequently used to capture smooth nonlinear temporal variation in epidemiological data. Random walk models assume that temporal effects evolve gradually over time rather than abruptly between consecutive periods (Blangiardo & Cameletti, 2015). The first-order random walk model (RW1) assumes that the temporal effect at time t depends on the effect at time $t-1$ plus random variation. RW1 models are particularly useful for modelling nonlinear climatic effects and long-term temporal trends where the relationship between environmental variables and disease transmission may vary smoothly over time (Blangiardo & Cameletti, 2015).

2.4.2. Spatial and space-time modelling

Spatial modelling is commonly used in disease mapping to account for the geographical dependence of disease occurrence between neighbouring areas. In spatial analysis, data are aggregated into a finite number of administrative units with clearly defined boundaries, such as districts or health regions (Morris et al., 2019). This is particularly relevant in disease mapping, because surveillance data are often reported at administrative levels rather than at individual household locations.

One challenge associated with areal disease data is that regions with small populations often produce unstable or noisy disease estimates (Morris et al., 2019). This is especially important for cholera because some districts may report very few cases during certain periods, while neighbouring districts may experience spikes in cases. Such cholera risk patterns may not have necessarily occurred that way, but rather arises from random variation due to small population sizes. Spatial models therefore attempt to reduce this instability by borrowing information from neighbouring regions, allowing the visible patterns of cholera risk.

Conditional Autoregressive (CAR) models are among the most widely used approaches for modelling spatial dependence in areal disease data. CAR models are based on the assumption that neighbouring regions tend to exhibit similar disease risks due to shared environmental, climatic, demographic, or socio-economic characteristics (Thiam et al., 2019). Because cholera can be transmitted through environmental and human pathways, neighbouring districts may share similar cholera risk due to common water bodies, sanitation conditions, transportation routes, climatic conditions, or population movement patterns.

To model spatial dependence in a Conditional Autoregressive (CAR) model, a neighbourhood structure is first specified to identify which regions are considered neighbours. Given a set of $\mathcal{N}$ spatial entities, two regions (i) and (j) are considered neighbours if they share a common border. This relationship is represented by the adjacency matrix w_{ij}, where, $w_{ij} = 1$ if two polygons are neighbours and $w_{ij} = 0$ otherwise. $w_{ii} = 0$ since a polygon cannot be a neighbour of itself (Morris et al., 2019).

One of the most commonly used spatial disease mapping models is the Besag York Mollié (BYM) (Besag et al., 1991). The BYM framework divides spatial variation in disease risk into two components: a spatially structured effect and a spatially unstructured effect (Blangiardo & Cameletti, 2015). The structured component is modelled using an Intrinsic Conditional Autoregressive (iCAR) process, which assumes that neighbouring regions may exhibit similar disease risks. By borrowing information from neighbouring areas, this component smooths risk estimates across space and may reduce instability caused dispersed case counts. The unstructured component captures residual variation that is independent of the neighbourhood structure. In the context of cholera, this may be localised outbreaks or district specific characteristics, such as differences in disease reporting, healthcare accessibility, environmental conditions, or other unmeasured factors that are not accounted for by the model covariates.

The observed cholera cases (Y_i) in i_{th} spatial unit are assumed to follow a Poisson distribution (MacNab, 2022):

$$Y_i \sim \text{Poisson}(\lambda_i) \quad \text{Equation 1}$$

$$\lambda_i = E_i r_i \quad \text{Equation 2}$$

where the average number of events λ_i, is equal to the expected number of cholera cases (E_i) in the i_{th} spatial unit, multiplied by the relative risk r_i, in each spatial unit. The logarithm of the relative risk is linked to a linear predictor η_i (Blangiardo & Cameletti, 2015):

$$\eta_i = log(r_i) = b_0 + u_i + v_i \quad \text{Equation 3}$$

where:

- b_0 = the overall intercept or average disease risk across all regions;
- u_i = the spatially structured random effect modelled using an Intrinsic Conditional Autoregressive (iCAR) process;
- v_i = the spatially unstructured random effect accounting for area independent heterogeneity in relative risk.

For disease mapping, the structured spatial effect is modelled using an iCAR prior, for mean vector $u = \{u_i, \dots, u_i\}$ with the conditional autoregressive distribution calculated as:

$$u_i|u_{-i} \sim \text{Normal}\left(\frac{1}{\mathcal{N}_i}\sum_{j=1}^{n} w_{ij}u_j, \frac{\sigma_u^2}{\mathcal{N}_i}\right) \quad \text{Equation 4}$$

where,

- u_{-i} = the vector of spatial effects for all regions except spatial unit i;
- $\mathcal{N}_i$ = The number of neighbours of spatial unit i;
- w_{ij} = is the adjacency indicator taking the value 1 when regions i and j are neighbours and 0 otherwise;
- σ_i^2 = variance for same area i which depends on the number of neighbours $\mathcal{N}_i$. This variance is small when the number of neighbours are many.

The unstructured random effect is assumed to be independently distributed as:

$$v_i \sim \text{Normal}(0, \sigma_v^2) \qquad \text{Equation 5}$$

where, σ_v^2 is the variance parameter that controls the variation amount between the spatially unstructured random effects (Blangiardo & Cameletti, 2015).

To investigate the influence of environmental and socioeconomic risk factors on cholera risk, the BYM model can be extended through ecological regression (Schrödle & Held, 2011). In this framework, covariates are incorporated into the linear predictor to explain part of the spatial variation in disease risk. The linear predictor can then be extended by adding $m = 1, \ldots, M$ number of covariates, such that:

$$\eta_i = b_0 + \beta_m x_{mi} + u_i + v_i \qquad \text{Equation 6}$$

where x_{mi} represents the covariates for each spatial unit i and β_m is the regression coefficient or fixed effects of spatial covariates, estimated by R-INLA.

Furthermore, the BYM model can be extended to a spatio-temporal framework through the inclusion of temporal random effects modelled using a random walk of order one (RW1) or order two (RW2) (Blangiardo & Cameletti, 2015). While the spatial component captures similarities in cholera risk between neighbouring districts, the temporal component captures changes in disease risk through time. Together, these components account for both spatial and temporal autocorrelation as well as variations unaccounted for by the correlation. In addition, RW1 and RW2 processes can be assigned to covariates to model potentially nonlinear relationships with cholera risk. This enables the model to capture gradual or complex changes in the effects of environmental and socioeconomic factors that may not be adequately represented by a linear assumption (Blangiardo & Cameletti, 2015).

2.4.3. Bayesian inference

While the space-time models mentioned above, provide a framework for representing spatial and spatio-temporal dependence, they are estimated within a Bayesian hierarchical modelling framework. Bayesian inference provides the statistical mechanism through which model parameters, including fixed effects, spatial random effects and temporal effects, are estimated. Consequently, understanding Bayesian inference is essential for interpreting how disease risk is estimated and uncertainty is quantified.

The major advantage of Bayes framework in epidemiological studies is its ability to learn and update its understanding. This involves combining prior information with the observed data to obtain marginal posterior distribution for each unknown model parameter (Blangiardo & Cameletti, 2015; Webb & Sidebotham, 2020). The resulting posterior distribution therefore represents the updated understanding of disease risk after the observed data is incorporated. This relationship is expressed through the Bayes' theorem:

$$p(\theta|y) \propto p(y|\theta) \times p(\theta) \qquad \text{Equation 7}$$

where $p(y|\theta)$ represents the likelihood and $p(\theta)$ the prior distribution. The resulting posterior distribution $p(\theta|y)$ summarises all available information about the parameters after observing the data (Webb & Sidebotham, 2020).

A major challenge in Bayesian analysis is the computational burden associated with estimating posterior distributions for complex models. Historically, Markov Chain Monte Carlo (MCMC) simulation methods were commonly used for Bayesian estimation. Although effective, MCMC approaches are computationally intensive and may become inefficient for large spatio-temporal datasets (Rue et al., 2009). To address these limitations, Integrated Nested Laplace Approximation (INLA) was developed as a computationally efficient alternative for latent Gaussian models. INLA approximates posterior distributions without relying on extensive simulation, hence reducing computation time while maintaining high levels of accuracy (Rue et al., 2009). Due to these advantages, INLA has been increasingly adopted in epidemiological and environmental health studies involving large spatial datasets and complex hierarchical structures (Letta et al., 2022). Bayesian model outputs are commonly interpreted using posterior summaries such as posterior means and credible intervals (Blangiardo & Cameletti, 2015). Credible intervals provide a probabilistic interpretation of parameter uncertainty and are frequently used in disease mapping studies to assess the strength and direction of associations between covariates and disease incidence (Farkhan et al., 2026a).

2.4.4. Model checking and selection

Model checking, also referred to as model criticism, is an important component of Bayesian disease mapping because it assesses whether a model provides a reasonable representation of the observed data and whether the assumptions underlying the model are appropriate. It can be used to evaluate the suitability of likelihood specifications, prior distributions, and model structures, as well as to determine whether the resulting posterior distributions are consistent with the observed data. Common approaches include cross validation, where a subset of the data is withheld for model evaluation, and posterior predictive checks, which compare observed data with data simulated from the fitted model. Other evaluation metrices include the root mean square error (RMSE) and R squared (R^2).

Model selection involves comparing competing models to identify the most appropriate representation of the underlying disease process. Several criteria have been proposed for this purpose, including the Deviance Information Criterion (DIC). Both criteria balance model fit against model complexity, thereby reducing the risk of overfitting while favouring models that adequately explain the observed data. In general, models with lower DIC and WAIC values are considered preferable and are commonly used in Bayesian spatial epidemiology to compare alternative model specifications.

The DIC is defined as (Ntirampeba et al., 2017):

$$\mathrm{DIC} = \overline{D} + p_D \qquad \text{Equation 8}$$

Where $\overline{D}$ represents the posterior mean deviance and P_D represents the effective number of parameters describing the model complexity (Ntirampeba et al., 2017).

3. METHODOLOGY

This chapter describes the methodological framework used to investigate the spatial and temporal dynamics of cholera risk in Cameroon. Covariates were first screened using Spearman correlation analysis and the Variance Inflation Factor (VIF) to identify potential associations with cholera incidence and assess multicollinearity among explanatory variables. The temporal dynamics of cholera transmission were then examined using an autoregressive process, followed by an assessment of the lagged nonlinear effects of hydroclimatic variables on cholera incidence. Finally, three Bayesian spatio-temporal disease mapping models were investigated to determine whether the inclusion of hydroclimatic, environmental, and socio-economic covariates explains the variation of cholera risk, followed by mapping of the residual risk distribution across Cameroon.

3.1. Study area

Cameroon is located in Central Africa and is administratively divided into 10 administrative regions. These regions are sub-divided to approximately 200 health districts. These are the operational scale units for disease surveillance and public health response. Often referred to as "Africa in miniature", the country encompasses a wide range of ecological and climatic zones, from the humid equatorial forests of the south to the semi-arid Saharan environment in the north (Mboringong et al., 2025). Cameroon experiences two seasons, rainy and dry seasons, with the intensity and duration varying sharply from the coastal and mountainous southwest to the arid northern lands (Mboringong et al., 2025). The terrain is highly diverse and includes the volcanic Cameroon Mountain Massif, extensive plateau regions, river basins, and low-lying flood-prone areas that influence water flow, sanitation conditions, and disease transmission dynamics. As of recent estimates, Cameroon has a population of approximately 27-28 million people.

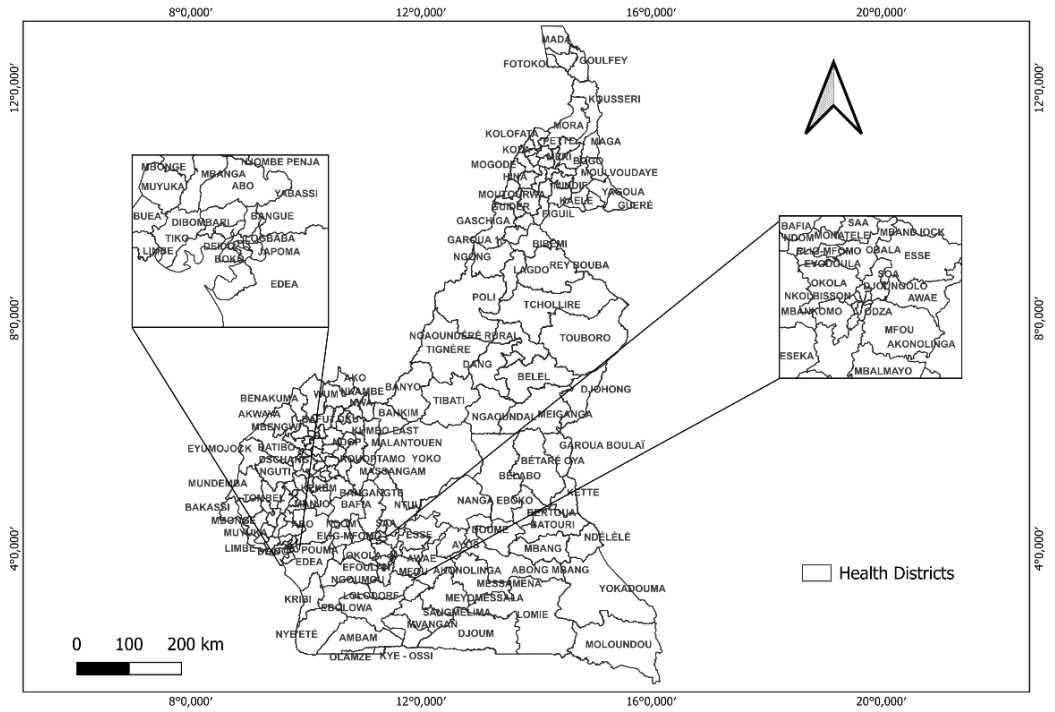


Figure 2: Map of Cameroon health districts

3.2. Flow chart

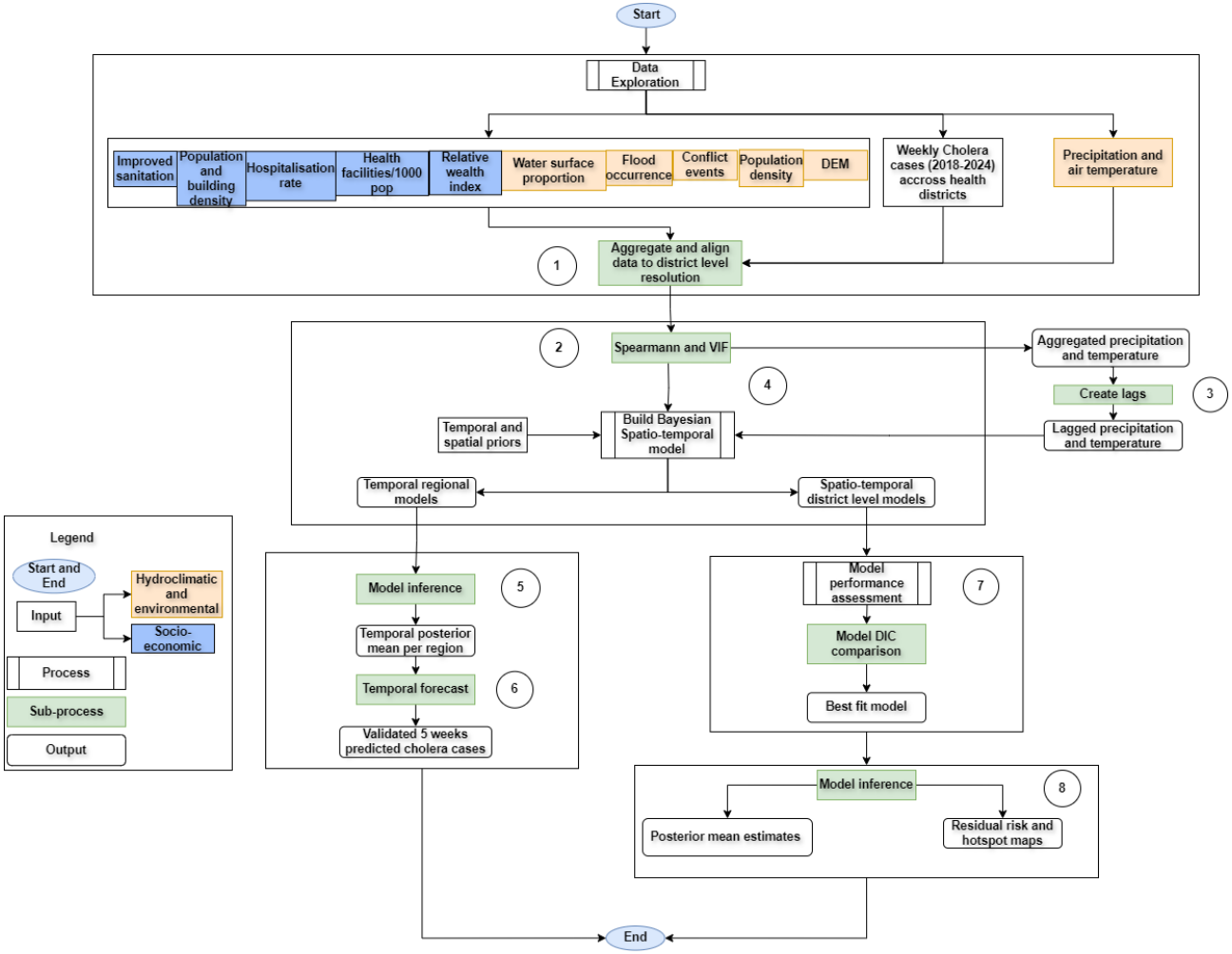


Figure 3: Flow chart

3.3. Dataset preparation

Cholera risk is conceptualised as the result of interactions between hydroclimatic conditions and underlying structural vulnerabilities. To represent these processes within the modelling framework, three broad classes of data were used: (1) cholera surveillance data, (2) climatic and environmental data, and (3) socioeconomic and vulnerability variables (Figure 3 section 1). cholera risk is conceptualised as an interaction of hazard and exposure, vulnerability, and coping capacity.

3.3.1. Cholera surveillance data

The outcome variable in this study is cholera incidence, observed at the health district level across Cameroon. National surveillance data were obtained from the Cameroon Ministry of Health through the national cholera monitoring system, under a non-disclosure agreement involving the Cameroon Red Cross and Netherlands Red Cross for research purposes. The dataset contains confirmed cholera cases originally recorded at the individual level in health facilities between 2017 and 2024. For this analysis, cases were aggregated into weekly counts per health district. This aggregation preserves patient confidentiality while maintaining sufficient temporal resolution to capture the rapid dynamics of cholera transmission, which often evolves within days to weeks following contamination events.

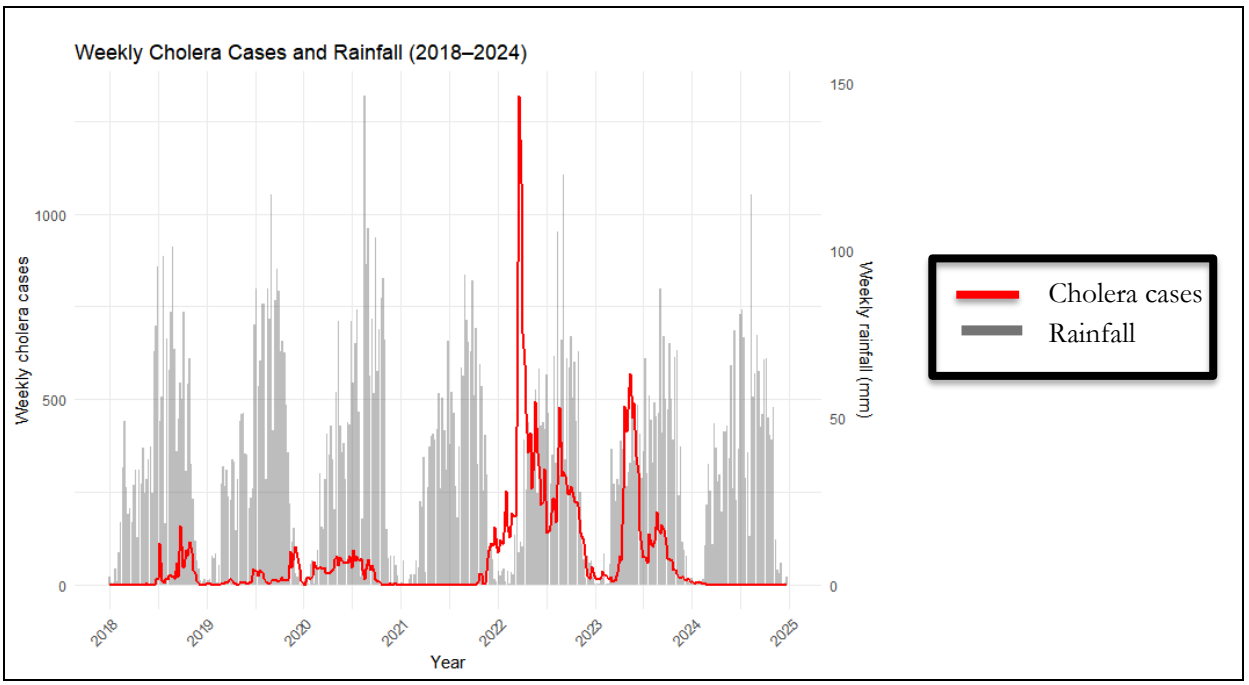


Figure 4: Weekly cholera cases (source: Cameroon Red Cross) and rainfall (source: Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) in Cameroon (2018–2024).

Prior to analysis, the cholera dataset underwent extensive cleaning to ensure consistency with spatial data. District names were harmonised through recoding to resolve inconsistencies in spelling, accents, punctuation, and administrative naming conventions between the epidemiological dataset and the spatial shapefile. Records with missing district identifiers were excluded, and the dataset was restricted to the study period (2017-2024). Weekly cholera case counts were then computed by grouping observations by year, epidemiological week, and district. To create a temporal structure for the temporal model, a continuous weekly time sequence (t_{week}) was generated and expanded across all districts, creating a full longitudinal district-weekly table. Weeks with no recorded cases were filled as zero cases, allowing for consistent time-series modelling across all regions and districts.

Spatial data were derived from a Cameroon health district shapefile provided by the Cameroon Red Cross. The shapefile was processed to extract and standardise district identifiers, remove duplicates, and assign unique numerical IDs for each health district and region. These identifiers are essential for spatial and hierarchical modelling approaches, such as Bayesian frameworks implemented in INLA. The cleaned spatial dataset was then joined with the complete district-weekly cholera dataset to produce the final modelling dataset. This integrated dataset includes epidemiological, temporal, and spatial attributes, forming the basis for subsequent spatio-temporal analysis of cholera risk across Cameroon.

3.3.2. Hydroclimatic data

3.3.2.1. Precipitation

Daily rainfall data for the period 2018 to 2023 were obtained from the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS), which provides quasi-global precipitation estimates at a spatial resolution of approximately 0.05° (~5 km). The rainfall data were downloaded as NetCDF raster files and processed in R using the terra package. The raster layers were reprojected to match the coordinate reference system of the health district boundaries before spatial extraction. Mean rainfall values were extracted for each health district using zonal statistics and aggregated to weekly totals and weekly averages (Figure 4). Weekly rainfall aggregation was used because cholera outbreaks usually occur after rainfall events within short delayed periods and this matched the temporal scale of the cholera cases (Lemaitre et al., 2019).

Rainfall lag effects were examined to account for the delayed influence of precipitation on cholera transmission. Heavy rainfall does not immediately result in cholera outbreaks, but may increase disease risk after a period which is required for environmental contamination followed human exposure (Reyburn et al., 2011). Consequently, lagged rainfall variables were included to capture the time interval between rainfall events and their potential impact on cholera incidence. Rainfall lags of up to four weeks were evaluated, consistent with previous studies that have reported delayed rainfall effects on cholera transmission ranging from approximately 15 to 30 days (Asadgol et al., 2020).

3.3.2.2. Flood occurrence

A binary flood occurrence variable was derived from weekly rainfall totals to capture extreme precipitation events likely to trigger flooding. District weeks with cumulative rainfall totals greater than or equal to 200 mm were classified as flood occurrence events and assigned a value of 1, while all other weeks were assigned 0. This was informed by the operational cholera trigger framework developed for Cameroon, which identifies elevated outbreak risk under conditions of prolonged intense rainfall, specifically when daily rainfall exceeds 50 mm for four consecutive days, and flooding affects more than 2,000 individuals (Red Cross Red Crescent Movement, 2025; Van Den Homberg et al., 2025).

The flood was also lagged to account for the delayed effects of flooding on cholera transmission. Flood events do not result in an immediate increase in reported cholera cases. Instead, flood runoffs goes through a pathway of contaminating drinking water sources through mobilisation of water pathogens, increasing human contact with contaminated surface water, damage water and sanitation infrastructure, and hence displace populations into overcrowded settings with limited access to safe water and hygiene facilities (Dadi et al., 2025).

3.3.2.3. Temperature

Temperature data were obtained from the ERA5 reanalysis dataset produced by the European Centre for Medium Range Weather Forecasts (ECMWF). Daily 2m air temperature raster data were extracted for each health district using the $extract()$ function from the terra package in R. Temperature values were converted from Kelvin to degrees Celsius and aggregated to weekly average temperature. Temperature was considered because warmer environmental conditions are associated with improved survival and growth of *V.cholerae* in aquatic environments. Temperature lag effects were evaluated for up to six weeks, as previous studies have shown that temperature may influence cholera transmission over longer temporal periods than rainfall (Asadgol et al., 2020; Dadi et al., 2025).

3.3.2.4. Digital Elevation Model (DEM)

Topographic characteristics were derived from the Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM) at 90 m spatial resolution. The elevation raster was reprojected to Universal Transverse Mercator (UTM) coordinates and slope was calculated in QGIS. Mean slope values were extracted for each health district using zonal statistics. In addition, the Topographic Wetness Index (TWI) was derived from the DEM using flow accumulation and slope raster to represent the potential for water accumulation within the landscape. Mean TWI values were similarly extracted for each health district and evaluated during covariate screening. Slope was included as an indicator of terrain steepness, which influences surface runoff and drainage processes. Flatter areas are generally more susceptible to water accumulation and flooding, which may increase cholera risk, while TWI was considered as an indicator of areas with greater potential for moisture retention and surface water accumulation (Diriba et al., 2026; Luque Fernandez et al., 2012).

3.3.2.5. Surface water proportion

Surface water proportion was derived from surface water data obtained from OpenStreetMap (OSM) through the Humanitarian Data Exchange (HDX). The data was reprojected to a projected coordinate system and intersected with the health district boundaries. The total area occupied by water bodies within each district was calculated and divided by the total district area to obtain the proportion of district area covered by surface water. This variable was included as an indicator of potential environmental exposure to contaminated water sources.

To represent population exposure to surface water, a proximity-based population metric was derived. Surface water polygons were buffered by 1 km and overlaid with a population density raster. The population residing within the buffered areas was estimated by summing raster values within each health district. This value was then divided by the total district population to obtain the proportion of the population living within 1 km of a surface water body. The resulting variable was used as an indicator of potential human exposure to contaminated water sources.

3.3.3. Socio-economic data

3.3.3.1. Population and population density

Population data between 2018-2023, were obtained from the WorldPop gridded population dataset at 100 m spatial resolution (Bondarenko et al., 2025) and aggregated to the health district level using zonal statistics. To improve consistency with the official population estimates for Cameroon, the aggregated WorldPop population projections were calibrated using figures reported in the "Population Cibles Prioritaires (2023)" report as a reference, while maintaining the temporal population growth trends represented in the WorldPop data. These population estimates were used to account for variation in district population sizes over time and to calculate the expected number of cholera cases across districts and years.

Population density data were obtained from the WorldPop gridded population density dataset for 2020 at approximately 1 km spatial resolution (WorldPop & Bondarenko, 2020). The raster data were aggregated to the health district level using zonal statistics, from which mean population density values were calculated for each district.

3.3.3.2. Building density

Built environment characteristics were represented using the Global Human Settlement Layer Built Surface (GHS BUILT-S) raster dataset developed by the European Commission Joint Research Centre (JRC). The dataset provides estimates of built-up surface area derived from Sentinel 2 and Landsat imagery. It was aggregated to the district level using zonal statistics with the exact_extract() function in R. Mean built up surface values were calculated for each district.

3.3.3.3. Hospitalisation rate

Hospitalisation rate was also derived from the cholera surveillance dataset to provide an additional indicator of outbreak severity and healthcare burden. Hospitalisation records were cleaned and recoded into binary outcomes indicating whether a cholera case was hospitalised. The weekly district level hospitalisation rates were then calculated as the proportion of hospitalised reported cases by the total reported case observations.

3.3.3.4. Poverty, healthcare facilities per 1000 population, unimproved sanitation and conflict

The Relative Wealth Index (RWI) dataset containing georeferenced point estimates of household wealth conditions across Cameroon was used. The mean district level RWI values were then calculated and converted into a poverty index by multiplying the values by negative one so that higher values represented greater poverty.

Data on healthcare facilities per 1000 population per health district was obtained from the Ministère De La Santé Publique (2023) report. This was calculated by dividing the total number of healthcare facilities in each district by the district total surface area (km^2) multiplied by 1000 population. Furthermore, Sanitation conditions were obtained from the World Health Organization Expanded Special Project for Elimination of Neglected Tropical Diseases (ESPEN). The dataset included district level estimates of access to improved sanitation facilities. The data was converted to a poor sanitation indicator by subtracting the improved sanitation mean by 1.

Data on conflict events was used to account for WASH disruptions and population vulnerability (Marou et al., 2024). The data was obtained from the Armed Conflict Location and Event Data Project (ACLED) and aggregated to the health district level. ACLED provides information on political violence, demonstrations, and conflict-related events, including the location, timing, actors involved, and reported fatalities associated with each event (Raleigh et al., 2023). Conflict records were linked to second level administrative units using administrative codes and subsequently joined to health district boundaries through spatial overlay. Weekly conflict events were then aggregated for each health district by summing the number of reported events occurring within each district and epidemiological week. To ensure temporal consistency with the cholera dataset, a complete district-week time series was created, and weeks with no reported conflict events were assigned a value of zero.

3.4. Data processing and explanatory analysis

3.4.1. Spearman correlation analysis and VIF check

To assess the initial relationships between the covariates and cholera cases and to identify potential multicollinearity among the explanatory variables, Spearman's rank correlation analysis was conducted (Figure 3 section 2). Spearman's correlation coefficient (ρ) measures the strength and direction of a monotonic relationship between two variables. Coefficient values range from -1 to 1, where negative values indicate an inverse relationship and positive values indicate a direct relationship. Values closer to -1 or 1 represent stronger associations and were considered an indication of potential multicollinearity between covariates. While values near 0 indicate weak or no association. Variance Inflation Factors (VIFs) were also calculated to further assess collinearity. VIF values greater than 10 were considered to be multicollinear with other covariates, whereas values below 10 were considered acceptable.

3.5. Temporal model analysis

This section evaluates the temporal autoregressive processes governing cholera transmission across Cameroon. The analysis aimed to determine whether weekly cholera incidence was influenced by incidence observed in preceding weeks and to assess the extent to which hydroclimatic factors contributed to the observed temporal dynamics. Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) analyses were used to identify an appropriate autoregressive structure for the weekly cholera time series.

3.5.1. Data table preparation

The temporal analysis was conducted using weekly cholera incidence data aggregated to the administrative regional level for the period 2018 to 2023. Regional aggregation was adopted to reduce sparsity in the weekly case series and to capture broad temporal patterns in cholera transmission. Weekly cholera case counts, population estimates, and average weekly rainfall and temperature were aggregated for each region.

Two regions, North West (REG_ID = 5) and Adamawa (REG_ID = 8), were excluded from the temporal analysis because no cholera cases were reported during the study period, resulting in unstable autoregressive model estimation. The analysis was therefore conducted for eight regions: Centre, East, Littoral, North, West, South West, Far North, and South.

A continuous weekly temporal index, t_k , was constructed independently for each region by chronologically ordering observations according to year and epidemiological week. This ensured that temporal dependence was estimated separately within each region rather than across a shared national time series.

Lagged rainfall and temperature variables were generated to investigate delayed hydroclimatic effects on cholera transmission. Rainfall lags of one, two and three weeks and temperature lags of one, two, three and four weeks were evaluated. These lagged variables were selected to account for delays between hydroclimatic conditions and their influence on cholera transmission. Continuous hydroclimatic variables were discretised into twenty groups using the $\text{inla.group}()$ function to facilitate flexible nonlinear modelling through random walk priors.

Prior to model fitting, the temporal structure of the weekly cholera incidence series was examined using the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF). The ACF was used to assess the correlation between cholera incidence and its previous values across multiple weekly lags, while the PACF was used to identify the dominant autoregressive order after accounting for intermediate lags. A

significant spike at lag one followed by a rapid decline in subsequent lags was interpreted as evidence supporting a first-order autoregressive process, AR(1). Based on these diagnostics, an AR(1) specification was adopted for all subsequent temporal analyses (GeeksforGeeks, 2023).

3.5.2. Model specification

This section specifies the model structure used to answer objective two. For each region k, a Bayesian temporal model was fitted using R-INLA package. The observed weekly cholera case counts for each region Y_{kt}, were assumed to follow a Poisson distribution. To account for the differences in the population at risk in each region, the expected cholera case counts, E_{kg} was included as an offset based on the national average risk in that year (Michal et al., 2025).

The expected cholera case count for each region was calculated based on national cholera incidence rate:

Therefore, for $i = 1, \dots, n$ data points aggregated into $k = 1, \dots, 10$ regions, $t = 1, \dots, 313$ weekly steps, and $g = 1, \dots, 6$ years:

$$E_{kg} = \left(\frac{\sum_{i=1}^{n} y_{ig}}{\sum_{i=1}^{n} P_g}\right) P_{kg} \qquad \text{Equation 9}$$

Table 1: Description of variables found in Equation 12

Variable	Description
E_{kg}	Expected number of cholera cases in region k, and year g
$\sum_{i}^{n} y_{ig}$	Total observed cholera cases in Cameroon during year g
P_{kg}	Population of region k during year g
$\sum_{i}^{n} P$	Total Cameroon population during year g

The baseline temporal model for region k at week t was defined as (Ebenezer et al., 2025; Ngwa et al., 2016):

$$y_{kt} \sim \text{Poisson}\left(E_{kg} \times r_{kt}\right) \qquad \text{Equation 10}$$

where r_{kt} is the rate or relative risk per region within week t. And the logarithm of the relative risk is connected to a linear predictor η_t for region k,

$$log(r_{kt}) = \eta_{kt} = \alpha_k + \omega_{kt} \qquad \text{Equation 11}$$

where:

- α_k = The model intercept per region k;
- ω_{kt} = The latent temporal process which changes over time with autoregressive method per region k.

The latent temporal process was modelled using a first-order autoregressive process supported by the ACF and PCF results. This process was defined as (Blangiardo & Cameletti, 2015):

$$\omega_{kt} = a\omega_{k(t-1)} + \epsilon_{kt} \quad \text{Equation 12}$$

where, for $t = 2, \dots, T$, the initial temporal effect is assumed to follow the stationary distribution, $\omega_{k1} \sim \text{Normal}\left(0, \sigma^2/(1-a^2)\right)$ (Gómez-Rubio, 2020). The term $\omega_{k(t-1)}$ is the structured temporal random effect for the same region k at the previous time point, $a \in (-1,1)$ is the autoregressive coefficient which describes the strength of temporal dependence between consecutive weeks, and ϵ_t is the zero mean Gaussian field assumed to be temporarily uncorrelated (Blangiardo & Cameletti, 2015). A value of a close to 1 indicates strong persistence in cholera transmission within consecutive weeks.

To evaluate the nonlinear effect of hydroclimatic factors on temporal cholera dynamics and lagged effects of cholera on itself, nonlinear rainfall and temperature effects were incorporated using first-order random walk (RW1) functions. Separate models were initially fitted for each lagged rainfall and temperature variable. These values were grouped into ordered into 20 groups using $inla.group()$ function in R INLA.

For each non-linear ordered groups of the covariate, $h = 2, \dots, \text{H}$,

$$f_k(x_h) = f_k(x_{h-1}) + \epsilon_{kh} \quad \text{Equation 13}$$

where

- $f_k(x_h)$ is the latent nonlinear effect associated with the h_{th} covariate group in region k,
- $f_k(x_{h-1})$ is the latent effect associated with the previous covariate group in same region,
- ϵ_h is an independent and identically distributed (iid) gaussian error with zero mean and precision τ.

The final autoregressive model with nonlinear hydroclimatic effects was defined as, for $i = 1, \dots, 3130$ data points aggregated into $k = 1, \dots, 10$ regions, $t = 1, \dots, 313$ weeks, and $g = 1, \dots, 6$ years:

$$\eta_{kt} = \alpha_{kt} + f(\text{Prec}_{kt}) + f(\text{Temp}_{kt}) + \omega_{kt} \quad \text{Equation 14}$$

where,

- α_{kt} = Model intercept per region k
- $f(.)$ = nonlinear rw1 effect
- $f(\text{Prec}_{kt})$ = nonlinear temporal trends of precipitation
- $f(\text{Temp}_{kt})$ = nonlinear temporal trends of temperature
- ω_{kt} = The latent spatio-temporal process which changes over time with autoregressive method per region k.

Model performance was evaluated using the DIC, and the best-performing lag for each variable and region was selected. Posterior means of the latent temporal effects ω_{kt} were extracted from the AR(1) component and compared between the baseline and hydroclimatic models. These temporal effects were retained on the linear predictor scale and were used to assess the extent to which rainfall and temperature explained temporal variation in cholera incidence.

3.5.3. Forecasting and Model validation

To assess the predictive performance of the temporal model, an out-of-sample validation approach was employed using a Poisson autoregressive model (Figure 3 section 6). Weekly cholera data from 2017 to

2022 was used to train the model. The first five epidemiological weeks of 2023 were withheld from model fitting and reserved for validation. Five future weekly observations were then added to each regional time series in the model, with cholera case counts for these five weeks specified as missing values (NA). Within the INLA framework, these missing values were treated as unknown quantities and predicted using the temporal dependence structure estimated from the training data.

To evaluate the forecasting accuracy of the model, the cumulative predicted and observed case counts for the first five epidemiological weeks of 2023 were compared. Only five weeks were used because the AR1 model relies on data from the immediate previous weeks to make predictions, therefore, the longer the forecasting period, the higher the prediction errors. The root mean square error (RMSE) was used for prediction validation. It quantifies the average magnitude of prediction errors, while placing greater weight on larger prediction errors. Hence, the residuals are squared before averaging between observed and predicted values within $\mathrm{n} = 1, \ldots, 5$ weeks and is defined as (Hodson, 2022):

$$RMSE = \sqrt{\frac{1}{n}\sum_{i}^{n}(y_i - \hat{y}_i)^2} \qquad \text{Equation 15}$$

where:

- n = the number of forecast weeks being validated
- y_i = observed cholera cases
- $\hat{y}_i$ = predicted cholera cases

Lower RMSE values indicate good predictive accuracy and strong agreement between observed and predicted values.

3.6. Spatial and spatio-temporal modelling

An ecological analysis was conducted to investigate the influence of fixed, nonlinear, and random effects on cholera transmission dynamics and the spatial distribution of cholera risk in Cameroon between 2018 and 2023. Bayesian disease mapping models were fitted using the Integrated Nested Laplace Approximation (INLA) framework implemented in R-INLA. Several model structures were developed and compared to assess their ability to explain and predict cholera risk across space and time.

3.6.1. Data table preparation

Weekly cholera surveillance data were aggregated at the health district level. The resulting dataset was organised in a longitudinal format, with each row representing a district-week observation and associated covariates (table 2). Unique identifiers were assigned to all model compartments, including years, weeks, regions, districts, and random effects, required by R-INLA model specification.

Continuous covariates modelled using random walk functions were partitioned into 20 groups using the $inla.group()$ function. This approach reduced computational complexity while allowing smooth nonlinear relationships between the covariates and cholera risk to be estimated (Gómez-Rubio, 2020).

To account for differences in population size across districts, expected cholera case counts were calculated using indirect standardisation. For each year, the national cholera incidence rate was obtained by dividing the total number of reported cholera cases by the total population. Expected counts for each district were then calculated by multiplying the district population by the corresponding annual national incidence rate. These expected counts were incorporated into the models as an offset.

Consider $i = 1, \dots, 62200$ observations aggregated into $j = 1, \dots, 200$ health districts, grouped into $k = 1, \dots, 10$ administrative regions, across $t = 1, \dots, 313$ weeks, and $g = 1, \dots, 6$ years. The expected number of cholera cases were calculated as:

$$E_{jkg} = \frac{\sum_i^n y_{ijktg}}{\sum_i^n P_{ijktg}} . P_{jkg} \qquad \text{Equation 16}$$

Table 2: Variables describing model parameters for Equation 20

Variable	Description
E_{jkg}	Expected number of cholera cases in district j, region k, during year g
y_{ijktg}	Observed cholera cases in district j, region k, during week t, and in year g
$\sum_i^n y_{ijktg}$	Total observed cholera cases in Cameroon in year g
P_{jktg}	Population of district j in region k, during year g
$\sum_i^n P_{ijktg}$	Total national population in Cameroon in year g

3.6.2. Standardised morbidity ratio

To provide an initial assessment of cholera risk across health districts, standardised morbidity ratio (SMR) were calculated. The SMR is the ratio of the observed number of cholera cases in a district and the number of cases that would be expected if the district experienced the same cholera incidence rate as the national population (Blangiardo & Cameletti, 2015; Naing, 2000).
Hence, the SMR was calculated as:

$$SMR_{jkg} = \frac{\sum_{i=1}^n Y_{ijkg}}{E_{jkg}} \qquad \text{Equation 17}$$

where $\sum_{i=1}^n Y_{ijg}$ is the sum of observed cholera cases in district j during year g

The SMR provides a relative measure of the baseline cholera risk by comparing observed cases to expected cases while accounting for differences in population size between districts. It indicates whether a district experiences a higher or lower burden of cholera when compared to the entire population. An SMR greater than 1 suggests a higher than expected cholera cases were observed, whereas an SMR less than 1 indicates a lower than expected risk of cholera

3.6.3. Model specification

The observed cholera case counts were assumed to follow a Poisson distribution:

$$y_{ijktg} \sim \text{Poisson}(E_{jkg} r_{ijktg}) \qquad \text{Equation 18}$$

where y_{ijktg} represents the observed number of cholera cases, E_{jkg} denotes the expected number of cases, and r_{ijktg} is the relative risk. The relative risk was linked to the linear predictor η_{ijktg}, through a logarithmic link function:

$$\log(r_{ijktg}) = \eta_{ijktg} \quad \text{Equation 19}$$

Spatial dependence was modelled using the Besag York Mollié (BYM) specification. The BYM model combines a structured spatial component that represents spatial autocorrelation between neighbouring districts with an unstructured spatial component which captures the district incidence variation not explained by spatial autocorrelation. Furthermore, the inclusion of spatial random effects helps account for overdispersion not fully explained by the Poisson likelihood (Morris et al., 2019).

A district adjacency matrix was derived from the Cameroon health district boundaries, whereby districts sharing a common border were defined as neighbours. The adjacency matrix was converted into an INLA graph object and used to specify the spatial component of the BYM model.

To investigate the spatio-temporal dynamics of cholera transmission and to identify a suitable representation of cholera risk, a series of Bayesian hierarchical models were specified. Model complexity was increased by progressively adding components to the models. The tested models are described below.

Model 1: Baseline Spatial model

The baseline model investigated the structured and unstructured spatial effects of districts using the BYM model. This model was conducted for each year, capturing the spatial variation between health districts and within the years. This model served as a reference against which the contribution of temporal and covariate effects can be evaluated.

$$\eta_{ijktg} = \alpha_g + U_{jg} + V_{jg} \quad \text{Equation 20}$$

where α_g is the year-specific intercept, U_{jg} and V_{jg} denotes the structured spatial effect and unstructured spatial effect respectively at district j and year g.

Model 2: Spatial and temporal random effects

Building on Model 1, structured and unstructured temporal random effects were incorporated to account for temporal variation in cholera transmission. The structured temporal component was modelled using a first-order random walk (RW1), which captures smooth changes in cholera risk through time. An independent and identically distributed (iid) temporal effect was additionally included to account for random temporal fluctuations not explained by the structured trend.

Although cholera incidence was analysed at the health district level, the structured temporal effect was estimated at the administrative regional level because districts within the same region experience broadly similar seasonal and climatic conditions, including rainfall and temperature patterns, which are important drivers of cholera transmission. Modelling a common regional temporal trend therefore allowed the model to capture shared climatic seasonality while the BYM spatial effect accounted for residual spatial heterogeneity between health districts. Although district-specific temporal interactions could capture more localised epidemic dynamics, they would increase model complexity

Although the analysis was conducted at the health district level, temporal effects were estimated at the administrative regional level. Estimating separate temporal trends for all 200 districts over 313 weeks would increase model complexity and computational burden. Grouping temporal effects at the regional

level allowed districts within the same region to share a common temporal pattern while maintaining spatial variation between districts through the BYM structure.

$$\eta_{ijktg} = \alpha_g + U_{jg} + V_{jg} + \gamma_{tk} + \phi_{tk} \quad \text{Equation 21}$$

where γ_{tk} represents the structured temporal effect modelled using RW1 and ϕ_{tk} represents the unstructured temporal effect modelled using an iid process

Model 3: spatio-temporal interaction model

To assess whether the inclusion of model covariates add information to the model in explaining variability, an ecological model to account for fixed and nonlinear effects of the covariates was developed. This model was used as the initial full model from which different covariate combinations were evaluated to identify the most appropriate model parameters.

Precipitation, temperature, and flood occurrence at a two-week lag were modelled as nonlinear effects using first-order random walk (RW1) functions to allow for potential non-linear relationships with cholera incidence. These effects were grouped at the regional level. The remaining covariates were included as fixed effects. The influence of each covariate was subsequently assessed through model comparison and statistical inference.

The initial model specification was:

$$\eta_{ijktg} = \alpha_g + f(\text{Prec}_{kt}) + f(\text{Temp}_{kt}) + f(\text{Flood}_{kt}) + \beta_1(\text{Slope}) + \beta_2(\text{Elevation}) + \beta_3(\text{TWI}) + \beta_4(PSW) + \beta_5(\text{PNW}) + \beta_6(\text{Pop_den}) + \beta_7(\text{Build_den}) + \beta_8(\text{Poor sanitation}) + \beta_9(\text{HCF}) + \beta_{10}(\text{PI}) + \beta_{11}(\text{Conflict}) + U_{jg} + V_{jg} + \gamma_{tk} + \phi_{tk} \quad \text{Equation 22}$$

Where $f(.)$ denotes nonlinear effects estimated using RW1 process and β represents fixed-effect regression coefficients.

Table 3: Covariates used in the spatio-temporal model

Variable	Description
Prec_{kt}	Average weekly precipitation varying across space k and time t included as nonlinear effect
Temp_{kt}	Average weekly temperature varying across space k and time t included as nonlinear effect
Flood_{kt}	Flood occurrence varying across time and space included as nonlinear effect
Pop_den	Population density constant over time
Build_den	Building density constant over time
Slope	Mean terrain constant over time
Elevation	Mean elevation
TWI	Mean Topographic wetness index
PSW	Proportion of surface water
PNW	Proportion of population within 1km buffer of a waterbody constant over time
PI	Mean poverty index over time
Poor sanitation	Percentage population with unimproved access to sanitation

	constant over time
Hcf	Healthcare facilities per 1000 population
Conflict	Total conflict events across time and space

3.6.4. Model comparison

The Deviance Information Criterion (DIC) was used to compare model performance between models (Figure 3 section 7). The model with the lowest DIC was considered the better fit model. comparing observed and predicted cholera cases through scatterplots, and the Pearson and spearman correlation tests, together with coefficients of determination, R squared (R^2) were additionally used to assess model predictive accuracy as shown in Appendix 4.

3.6.5. Model inference

The best fitted model based on lowest DIC was selected for posterior inferences on the model parameters (Figure 3 section 8). These included posterior means and 95% credible intervals extracted for all fixed and non-linear covariate effects, and summaries of spatial random effects and temporal random effects.

For fixed-effect covariates, posterior mean estimates were used to quantify the direction and magnitude of association between the explanatory variables and cholera risk. The 95% credible interval was also assessed, where intervals not containing zero were considered a significant evidence of association (Farkhan et al., 2026). For nonlinear covariates, posterior mean estimates and 95% credible intervals were examined to assess potential non-linear relationships between the covariates and cholera incidence. For interpretation, posterior estimates of fixed effects were exponentiated to obtain relative risks (RR). RR values higher than one indicated an increase in cholera risk was associated with an increase in the covariate, whereas values less than one indicate a reduction in risk.

The posterior distributions of the structured and unstructured temporal effects were also examined to assess temporal trends in cholera transmission across the administrative regions. Similarly, posterior spatial random effects were evaluated to identify the proportion explained by the spatially structured variation in cholera risk after accounting for the included covariates and temporal effects. This proportion of variation is determined by computing the spatial fraction, $\text{frac}_{spatial}$, variance.

Considering n for total number of health districts, let u_i represent the spatially structured residual for health district $j = 1, \ldots, 200$. The posterior marginal variance for the structured spatial effect was calculated as:

$$s_u^2 = \frac{\sum_{j=1}^{n}(u_j - \bar{u})^2}{n-1} \quad \text{Equation 23}$$

$$\text{frac}_{spatial} = \frac{s_u^2}{s_u^2 + \sigma_v^2} \quad \text{Equation 24}$$

where,

- s_u^2 = posterior marginal variance for the structured spatial effect (CAR component)
- u_i = spatially structured residual
- $\bar{u}$ = mean of structured spatial residual
- σ_v^2 = marginal variance for the unstructured spatial effect (iid component)

3.7. Relative risk and hot-spot mapping

To assess the spatial distribution of cholera residual risk across Cameroon after accounting for covariates and temporal random effect, districts level posterior relative risks were derived from the fitted Bayesian spatio-temporal model. The district specific relative risk r_{jg}, was extracted from the BYM spatial effect from Models 2 and 3:

$$r_{jg} = exp(U_{jg} + V_{jg}) \quad \text{Equation 25}$$

Furthermore, posterior exceedance probabilities were employed to identify districts with elevated or lower residual spatial risk. Exceedance probabilities is a probabilistic measure of uncertainty, by quantifying the posterior probability that the relative risk in each district exceeds a specified threshold (Farkhan et al., 2026). This is contrary to the frequentist approach which uses p-values or confidence intervals to assess statistical evidence (Blangiardo & Cameletti, 2015).

A relative risk threshold of one was adopted because a relative risk greater than one indicates that the disease risk in a health district exceeds the overall expected risk, whereas a relative risk less than one indicates lower than expected risk. Since the latent BYM spatial effect is estimated on the logarithmic scale, the probability that the residual relative risk exceeds one is equivalent to the probability that the latent spatial effect exceeds zero. Therefore, the exceedance probability was defined as:

$$Pr(r_{jg} > 1|y) = \Pr(U_{jg} + V_{jg} > 0|y) \quad \text{Equation 26}$$

Where r_{jg} is the posterior relative risk for the spatial random effect, and y is the observed cholera case counts. Posterior exceedance probabilities were calculated from the marginal posterior distributions estimated by INLA:

$$Pr(r_{jg} > 1|y) = 1 - \Pr(r_{jg} \leq 1|y) \quad \text{Equation 27}$$

Several studies have explored different probability of exceedance threshold ranging from 0.7 to 0.95 for hotspot identification (Farkhan et al., 2026; Richardson et al., 2004; Rotejanaprasert et al., 2024). In this study, districts with exceedance probabilities greater than 0.8 (1-α with α = 0.05) were classified as hotspots, indicating strong evidence that cholera risk exceeded the national average. Following Richardson et al. (2004), districts with posterior exceedance probabilities of at least 0.80 were classified as residual hotspots, as this cutoff provides a good balance between sensitivity and specificity for identifying areas of elevated disease risk. Conversely, districts with exceedance probabilities below 0.2 were considered low-risk areas. The resulting hotspot maps were used to identify areas with persistently elevated cholera risk and to assess spatial uncertainty in the modelled risk estimates.

4. RESULTS

This chapter presents the results of the study, beginning with descriptive statistics and an assessment of relationships among the explanatory variables using correlation and variance inflation factor analyses. Regional autoregressive patterns in cholera incidence are then examined, followed by Bayesian spatio-temporal modelling to investigate the spatial and temporal distribution of cholera risk and the influence of selected covariates across Cameroon.

4.1. Exploratory epidemiology analysis

Before addressing the specific objectives, an exploratory analysis of cholera incidence and associated covariates across health districts in Cameroon between 2019 and 2023. The objective of the exploratory analysis was to examine the distribution of the outcome and explanatory variables and identify potential spatial and temporal patterns in cholera incidence. The analysis was conducted using weekly cholera case counts aggregated at the health district level. Descriptive summaries were first computed for all variables used in the study (Table 4). The covariates showed variability across districts and across time.

4.1.1. Dataset summary statistics

A total of 25,475 cholera cases were recorded in Cameroon between 2018 and 2023. Annual case counts remained relatively low between 2018 and 2021, ranging from 727 to 1,890 cases per year. A spike in cholera incidence occurred in 2022, when 14,460 cases were reported, representing the largest outbreak during the study period. Table 4 presents the summary statistics of the variables included in the analysis. Cholera cases were generally low across district weeks, with a mean of 0.31 cases and a median of zero, indicating that most districts reported no cases during many weeks. Weekly rainfall ranged from zero to 57.81 mm and temperatures ranged from 16.88°C to 38.1°C across health districts.

Population density ranged from 3.71 to 23,831 persons/km², while unimproved sanitation levels were generally high, with a mean value of 0.53. Hence 53% of the population in the district had poor sanitation. The proportion of the population living close to water bodies ranged from 0 to 1, indicating that some districts had no population living within 1km proximity to a waterbody, while others had their entire population living within 1km proximity to a waterbody respectively.

Table 4: Variables summary statistics

Variable	Mean	SD	Median	Min	Max
Cholera cases	0.31	4.71	0	0	860
SMR	0.02	0.77	0	0	98.75
Average rainfall	4.88	5.1	3.96	0	57.81
Average temperature	24.84	2.98	24.58	16.88	36.39
Population	133294	115069.48	99529	13527	640608
Population density	936.4	3204.89	87.91	3.71	23830.9
Elevation	667.27	412.65	634.21	10.37	1954.64
Slope	0.09	0.06	0.07	0.004	0.32
TWI	8.59	1.08	8.45	6.61	12.56
Surface water proportion	0.05	0.12	0	0	0.88
Proportion population within 1km buffer to waterbody	0.26	0.28	0.14	0	1
Flood occurrence	0	0.06	0	0	1
Building density	0.1	0.2	0.01	0	0.98
Hospitalisation rate	36.78	178.04	0	0	1000
Health facilities per 1000 pop	0.26	0.17	0.23	0.01	0.99
Poverty index	0.17	0.39	0.3	-1.57	0.7
Unimproved sanitation (%)	0.53	0.1	0.54	0.23	0.92
Conflict events	0.2	2.18	0	0	87

4.1.2. Collinearity check with Spearman rank correlation and VIF

Spearman rank correlation analysis was conducted to assess the relationships between cholera incidence and the explanatory variables, as well as to identify potential multicollinearity among covariates. The correlation matrix is presented in Figure 5.

Most covariates exhibited weak correlations with cholera case counts, suggesting limited association with the response variable during exploratory analysis. Population density $\rho = 0.21$ and building density $\rho = 0.22$ showed weak positive correlations with cholera incidence. In contrast, unimproved sanitation $\rho = -0.08$, slope $\rho = -0.10$, and healthcare facilities per 1000 population $\rho = -0.01$, showed weak negative associations with cholera incidence.

Among the explanatory variables, a strong positive correlation was observed between population density and building density $\rho = 0.87$, indicating major overlap between the two variables. Both variables represent characteristics associated with human settlement concentration and urbanisation, suggesting the presence of multicollinearity. A strong negative correlation was also observed between building density and poverty index $\rho = -0.81$, while temperature and slope exhibited a moderate negative correlation $\rho = -0.64$ and positive correlation with TWI $\rho = 0.66$. While slope and TWI exhibited multicollinearity with strong negative correlation $\rho = -0.94$.

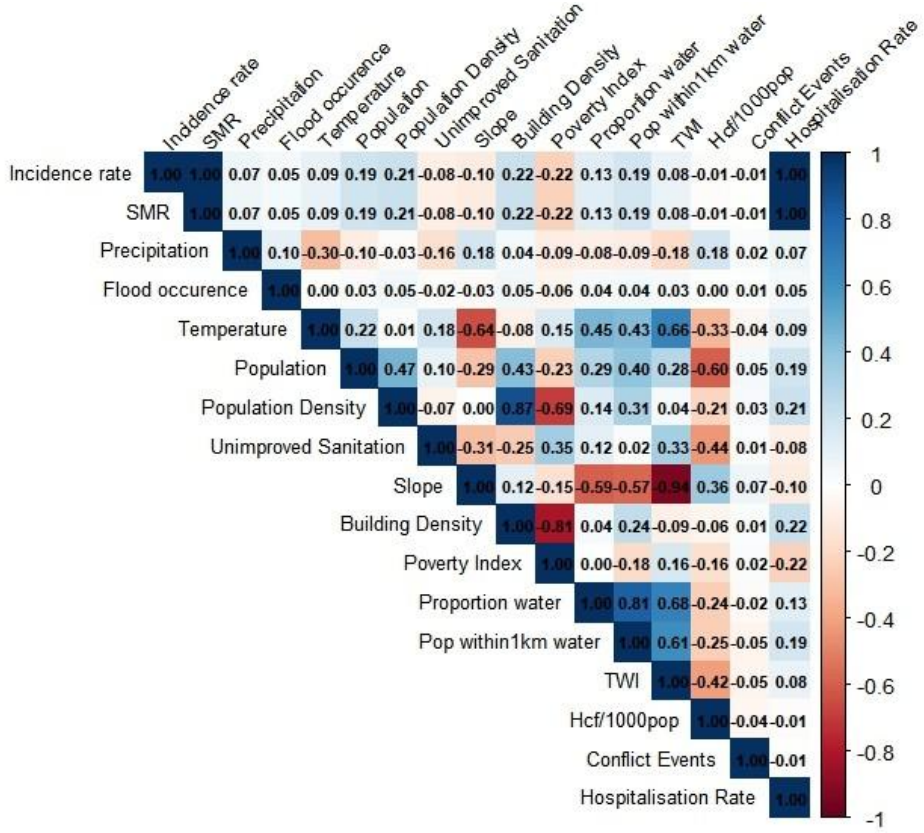


Figure 5: Spearman correlation matrix illustrating the relationships between cholera incidence and explanatory variables and the interrelationships among the explanatory variables

To further evaluate multicollinearity among the predictor variables, the Variance Inflation Factor (VIF) was calculated for each covariate. VIF values greater than 10 were considered indicative of severe multicollinearity. The VIF results are presented in Table 5. Building density had a VIF value of 12.49, exceeding the acceptable threshold and indicating high collinearity with other explanatory variables. This finding was consistent with the correlation analysis, which showed strong associations between building density, population density, and poverty index. Consequently, building density was excluded from subsequent modelling to reduce redundancy and improve model stability.

All remaining covariates exhibited VIF values below the recommended threshold and were therefore used for further analysis. These variables were incorporated into the Bayesian spatio-temporal models, and different covariate combinations were subsequently evaluated. The model with the lowest DIC was selected as the best fit model specification.

Table 5: Variance Inflation Factor

Variables	VIF value	Interpretation
Average precipitation	1.25	Acceptable
Average Temperature	1.67	Acceptable
Flood event occurrence	1.12	Acceptable
Population density	4.30	Acceptable
Unimproved sanitation (%)	1.32	Acceptable
Slope	3.77	Acceptable
TWI	5.4	Acceptable
Elevation	2.78	Acceptable
Building density	12.49	High collinearity
Poverty	8.27	Moderate collinearity
Surface water proportion	2.0	Acceptable
Population within 1km water	2.9	Acceptable
Health facilities per 1000 pop	1.86	Acceptable
Number of conflict events	1.01	Acceptable
Hospitalisation rate	1.17	Acceptable

4.2. Objective 1: To evaluate the temporal autoregressive process governing cholera transmission across time.

The temporal autoregressive process aimed to determine whether weekly cholera incidence was influenced by incidence observed in previous weeks and evaluate the effect of hydroclimatic factors on the unobserved temporal dynamics.

4.2.1. Identification of temporal dependence

Temporal autocorrelation analysis was conducted to evaluate the temporal dependence of cholera transmission across regions in Cameroon. The autocorrelation function (ACF) and partial autocorrelation function (PACF) were examined for each region to identify the most appropriate temporal modelling structure for the cholera time series.

A positive ACF was observed across all regions (Appendix 1), indicating that cholera incidence in a given week was associated with incidence in previous weeks. The PACF plots provided further evidence of direct temporal autocorrelation between consecutive weeks. In most regions, the PACF exhibited a spike at the first lag followed by a rapid decline, suggesting that much of the temporal structure could be explained by a first order autoregressive process, AR(1). Based on these observations, an AR(1) model was selected to characterise temporal dependence in weekly cholera incidence between 2018 and 2023.

4.2.2. Regional temporal dependence

Total cholera cases between 2018 to 2023 varied across regions ranging from 0 to 9993 cases (Table 6). Littoral recorded the highest cumulative burden with 9,993 reported cases, followed by South West with 6,617 cases and Centre with 5,576 cases. In contrast, East reported only 20 cases throughout the study period, while no cases were recorded in North West and Adamawa.

The AR(1) correlation coefficients ρ, further confirmed the presence of strong temporal persistence in cholera incidence across Cameroon. Estimated ρ values ranged from 0.89 in the East region to 0.98 in, North region.

Table 6: Region total cases and time dependence coefficient across study period (2018-2023)

Region ID	Region	Total cases	ρ (AR1)
1	Centre	5576	0.92
2	East	20	0.89
3	Littoral	9993	0.97
4	North	1223	0.98
5	North West	0	-
6	West	271	0.97
7	South West	6617	0.96
8	Adamawa	0	-
9	Far North	739	0.92
10	South	1058	0.96

4.2.3. Influence of hydroclimatic variables on the temporal process

To evaluate the lagged effects of cholera incidence on itself and the nonlinear influence of hydroclimatic conditions on temporal cholera dynamics, lagged rainfall and temperature variables were incorporated into an AR(1) modelling framework using first-order random walk (RW1) functions. Multiple lag structures were evaluated for each region, and the rainfall and temperature lags selected as best lags per region are presented in Appendix 2.

Figure 6 compares the posterior mean temporal effects estimated from the baseline AR(1) model and the hydroclimatic AR(1) model across the study regions. In general, both models reproduced similar temporal patterns, indicating that temporal persistence remained the dominant driver of cholera dynamics. Differences between the baseline AR(1) model and the hydroclimatic AR(1) model were observed in several regions. The regions with observed difference between both temporal processes, were the North, South, South west, and West regions. In these regions, the hydroclimatic model reduced the magnitude of the temporal effect during major outbreak periods, indicating that part of the temporal variability previously captured by the autoregressive process could be explained by lagged rainfall and temperature conditions.

In contrast, the temporal processes estimated for Centre, Littoral, East and Far North were almost identical between the baseline and hydroclimatic models. This indicated that while hydroclimatic conditions contributed to temporal variation in some regions, much of the observed persistence in cholera transmission was not explained by rainfall and temperature alone.

4.2.4. Forecast validation

The predictive performance of the hydroclimatic AR(1) model was evaluated by fitting the model using data from 2018 to 2022 and forecasting cholera incidence for the first five epidemiological weeks of 2023 and visualised in Appendix 3.. Forecast accuracy was assessed using the root mean square error (RMSE) between observed and predicted weekly case counts (Table 7).

Table 7: Forecast validation results for the combined AR(1) model incorporating lagged hydroclimatic covariates.

Region	**Total observed cases**	**Total predicted cases**	**RMSE**
Centre	35	100.69	13.38
East	0	5.04	1.10
Littoral	65	325.77	60.28
North	0	8.31	1.81
West	5	6.47	0.75
South West	0	4.77	1.05
Far North	0	9.21	2.07
South	0	9.45	2.03

Regions with very low cholera incidence, including East, North, South West, Far North and South, exhibited relatively small forecast errors, with RMSE values ranging from 1.10 to 2.07 cases. In contrast, forecast accuracy was lower in regions with larger cholera burdens. Littoral exhibited the highest prediction error, with an RMSE of 60.28 cases. During the five-week validation period, the model predicted a total of 325.77 cases compared with 65 observed cases. Predictions for the Centre region had a relatively high forecast error, with an RMSE of 13.38 and a total predicted count of 100.69 cases compared with 35 observed cases.

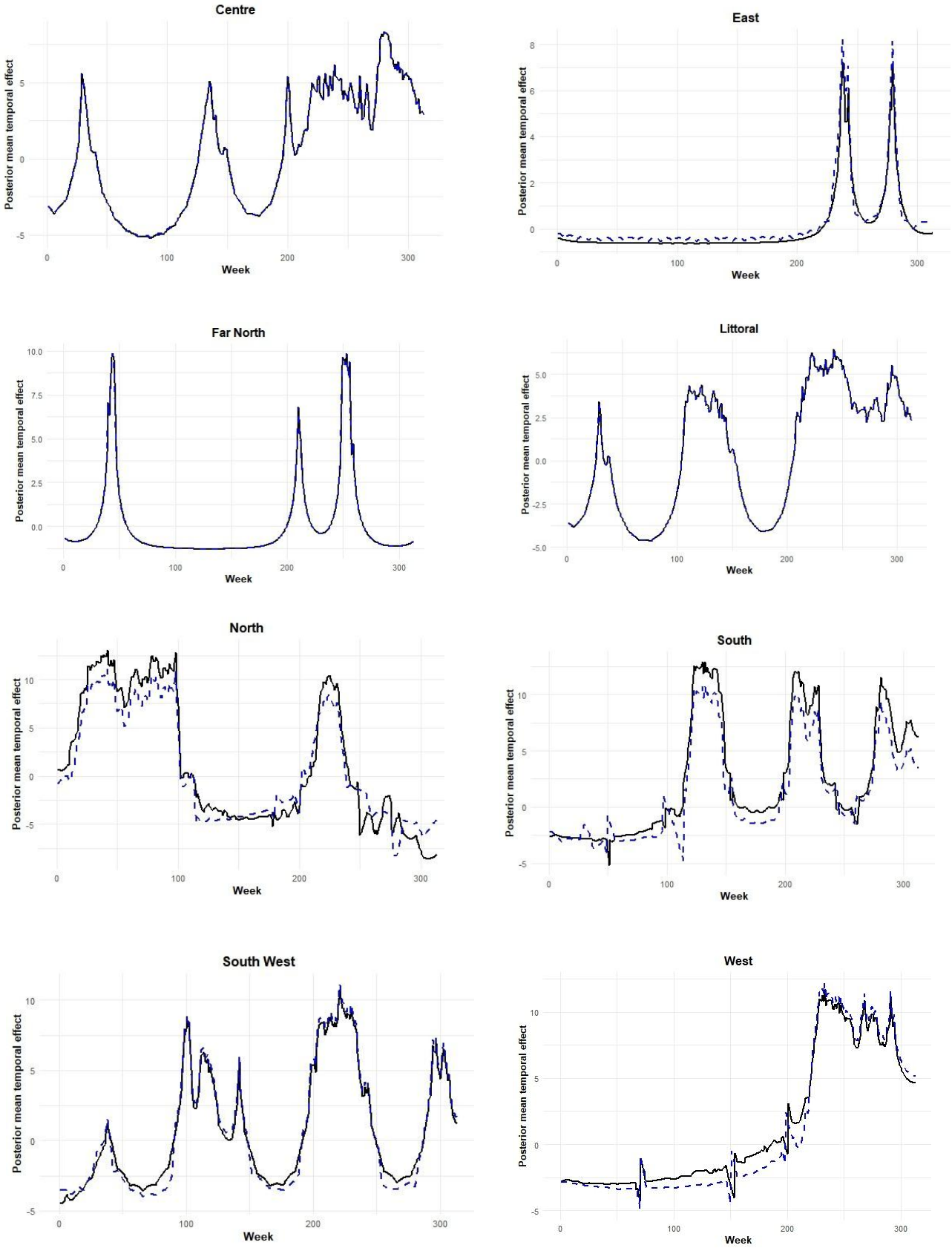


Figure 6: Posterior mean temporal effects from the baseline AR(1) model (solid line) and hydroclimatic AR(1) model (dashed line) for each study region between 2018 and 2023.

4.3. Objective 2: To assess the effects of selected hydroclimatic, environmental, and socio-economic covariates on the spatio-temporal distribution of cholera incidence.

Bayesian spatio-temporal models were fitted to assess cholera risk patterns across Cameroon between 2018 and 2023 while accounting for spatial, temporal, and explanatory variables effects. Model performance was evaluated using the Deviance Information Criterion (DIC), with lower values indicating improved model fit (Table 8).

4.3.1. Standardised Morbidity Ratio

Figure 4 presents the spatio-temporal distribution of the Standardised Morbidity Ratio (SMR) across Cameroon between 2018 and 2023. Districts with an SMR greater than one experienced cholera incidence above the national average, whereas districts with an SMR below one had lower than average incidence.

Shifts in the spatial distribution of cholera risk were observed throughout the study period. Between 2018 and 2019, districts with elevated SMRs were mainly concentrated in the North and Far North regions. From 2020 onwards, higher-risk districts increasingly were observed in the southern and western parts of the country, particularly within the South-West, Littoral, Centre, and neighbouring regions. The largest number of districts with cholera risk above one occurred in 2022 and 2023, with 25 and 26 districts, respectively, recording SMR values greater than one.

Spatial clustering of elevated SMRs was evident in several years, particularly where neighbouring districts exhibited similar levels of cholera risk. This pattern suggests that cholera incidence was not randomly distributed across space and provides preliminary evidence of spatial dependence in disease occurrence. These findings support the use of spatial modelling approaches that explicitly account for spatial autocorrelation when estimating cholera risk.

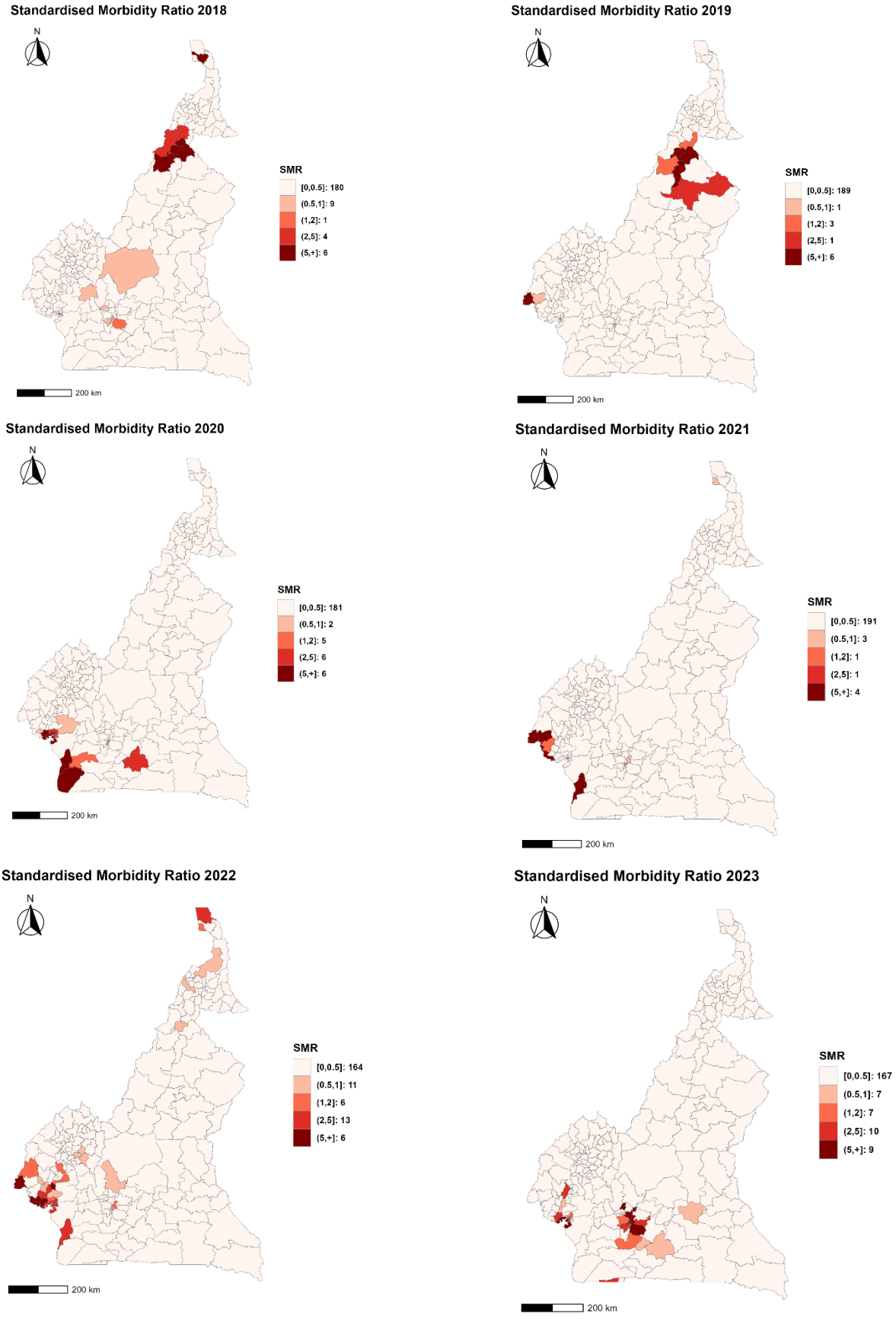


Figure 7: Annual Standardised Morbidity Ratio (SMR) maps showing the spatial distribution of cholera risk across health districts in Cameroon between 2018 and 2023.

4.3.2. Model comparison

The baseline model containing only spatial random effects produced the highest DIC value, suggesting that cholera incidence could not be adequately explained without accounting for temporal dependence. Model performance improved with the inclusion of temporal random effects. The addition of hydroclimatic, environmental, and socio-economic covariates, including average weekly rainfall, average weekly temperature, water surface proportion, slope, unimproved sanitation, and poverty index, resulted in a further reduction in DIC values.

Model 3 was identified as the best-performing spatio-temporal model, with the lowest DIC value among the tested models (Table 8). The model was comprised of BYM spatial random effects, a first-order random walk (RW1) temporal structure, and both fixed and nonlinear covariate effects. The covariates were selected based whether adding the covariate increased or decreased Model 3 DIC.

The final Model 3 specification was:

$$\eta_{ijktg} = \alpha_g + f(\text{Prec}_{kt}) + f(\text{Temp}_{kt}) + \beta_1(\text{Slope}) + \beta_2(\text{PSW}) + \beta_3(\text{Poor sanitation}) + \beta_4(\text{Poverty index}) + \beta_5(\text{Hcf}) + U_{jg} + V_{jg} + \gamma_{tk} + \phi_{tk} \quad \text{Equation 28}$$

where,

- α_{jg} = Model intercept
- Prec_{kt} = Rainfall lagged by two weeks
- Temp_{kt} = Temperature lagged by two weeks
- Slope = Percentage slope per district
- PSW = Proportion of surface water in each district
- Poor sanitation = Unimproved sanitation per district
- Poverty index = Poverty index per district
- Hcf = Healthcare facilities per 1000 population
- U_{jg} = Spatially structured random effect modelled using BYM2
- V_{jg} = Spatially unstructured random effect
- γ_{kt} = Structured temporal effect (RW1) grouped by administrative region
- ϕ_{tk} = Unstructured temporal effect (iid model)

Table 8: Model performance comparison, 2018-2023 (numbers in bold indicate the lowest DIC results)

Models	DIC
Model 1	47434.84
Model 2	30111.12
Model 3 with Equation 28 model specification	**29397.02**

4.3.3. Model fit of observed and predicted cholera cases.

Appendix 4 represents the temporal fit of the Model 3 predicted values, to the observed weekly cholera cases between 2018 and 2023. Overall, the predicted values closely fitted the observed cholera trends across the regions. Hence, the model captured the temporal dynamics and outbreak patterns adequately.

4.3.4. Fixed and non-linear effects

Fixed effects were inferred from the best fitting model (Model 3) and are presented in Table 9. Cholera incidence was negatively associated with both the poverty index and unimproved sanitation. The poverty

index showed a statistically significant negative association with cholera incidence (β = -1.459, 95% CrI: (-2.417, -0.1489), indicating lower cholera risk in districts with higher poverty index values. Unimproved sanitation was also negatively associated with cholera incidence (β = -0.059, 95% CrI: (-3.154, 3.028); however, the wide credible interval indicates high uncertainty around this estimate.

Percentage slope exhibited a significant negative association with cholera incidence (β = -0.112, 95% CrI: (-0.201, -0.024), suggesting that districts characterised by steeper terrain had lower cholera risk. Conversely, lower-lying areas may be more susceptible to water accumulation and environmental conditions that facilitate cholera transmission.

The proportion of water bodies within a health district and the number of healthcare facilities per 1,000 population were positively associated with cholera incidence, with posterior mean β = 2.822 (95% CrI: 0.526, 5.112) and 0.884 (95% CrI: (-1.180, 2.963), respectively.

Table 9: Posterior mean regression coefficients (β) and 95% credible intervals (CrI) for the fixed effects included in the best fitting Bayesian spatio-temporal cholera model, Model 3 (Bold values indicate covariates whose 95% credible intervals for posterior mean do not include the null value.

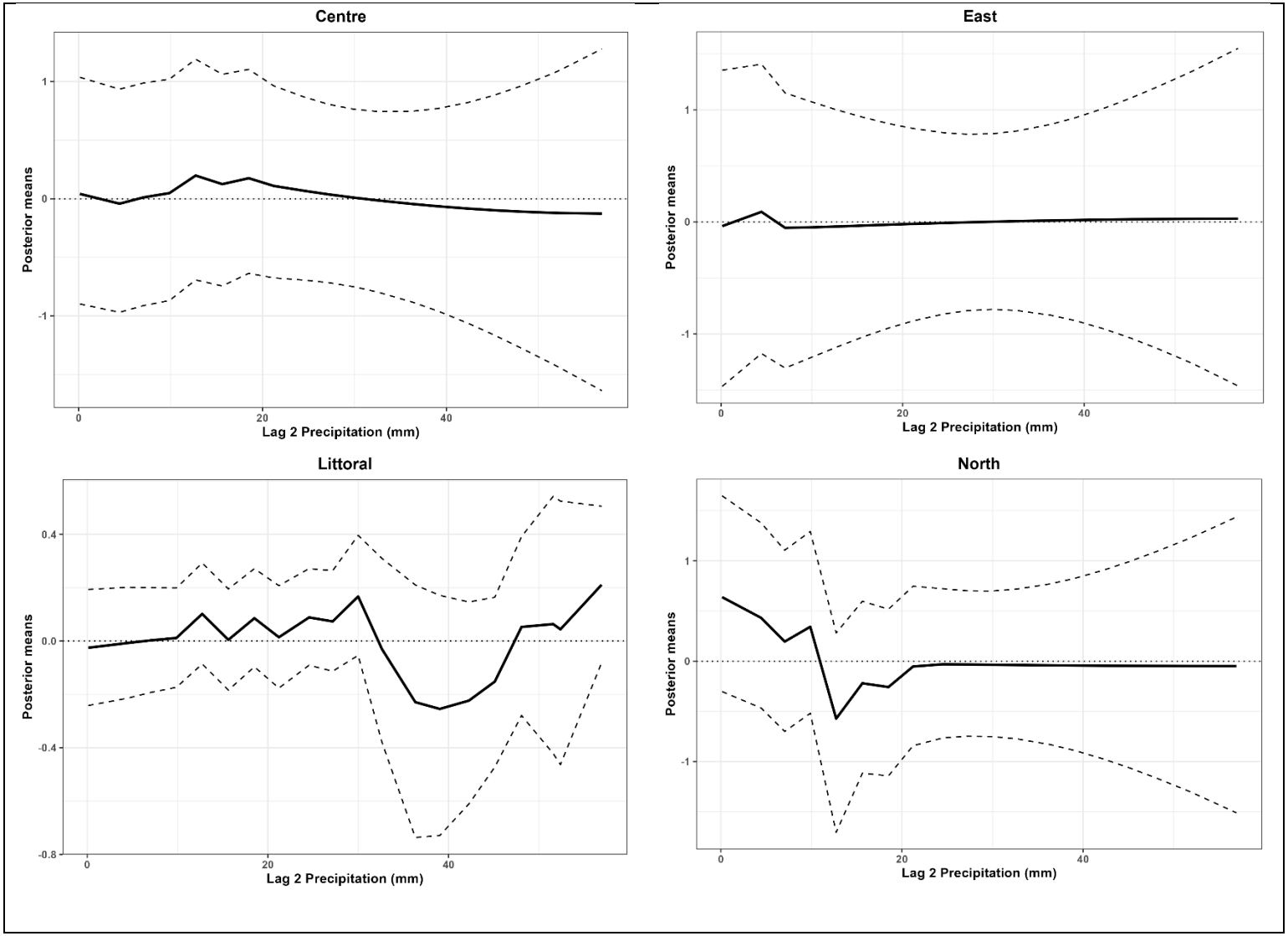
Covariate	β (posterior mean)	95% CrI
Unimproved sanitation	-0.059	(-3.154, 3.028)
Poverty Index	**-1.459**	**(-2.417, -0.489)**
Percentage Slope	**-0.112**	**(-0.201, -0.024)**
Water surface proportion	**2.822**	**(0.526, 5.112)**
Healthcare facilities per 1000	0.884	(-1.180, 2.963)

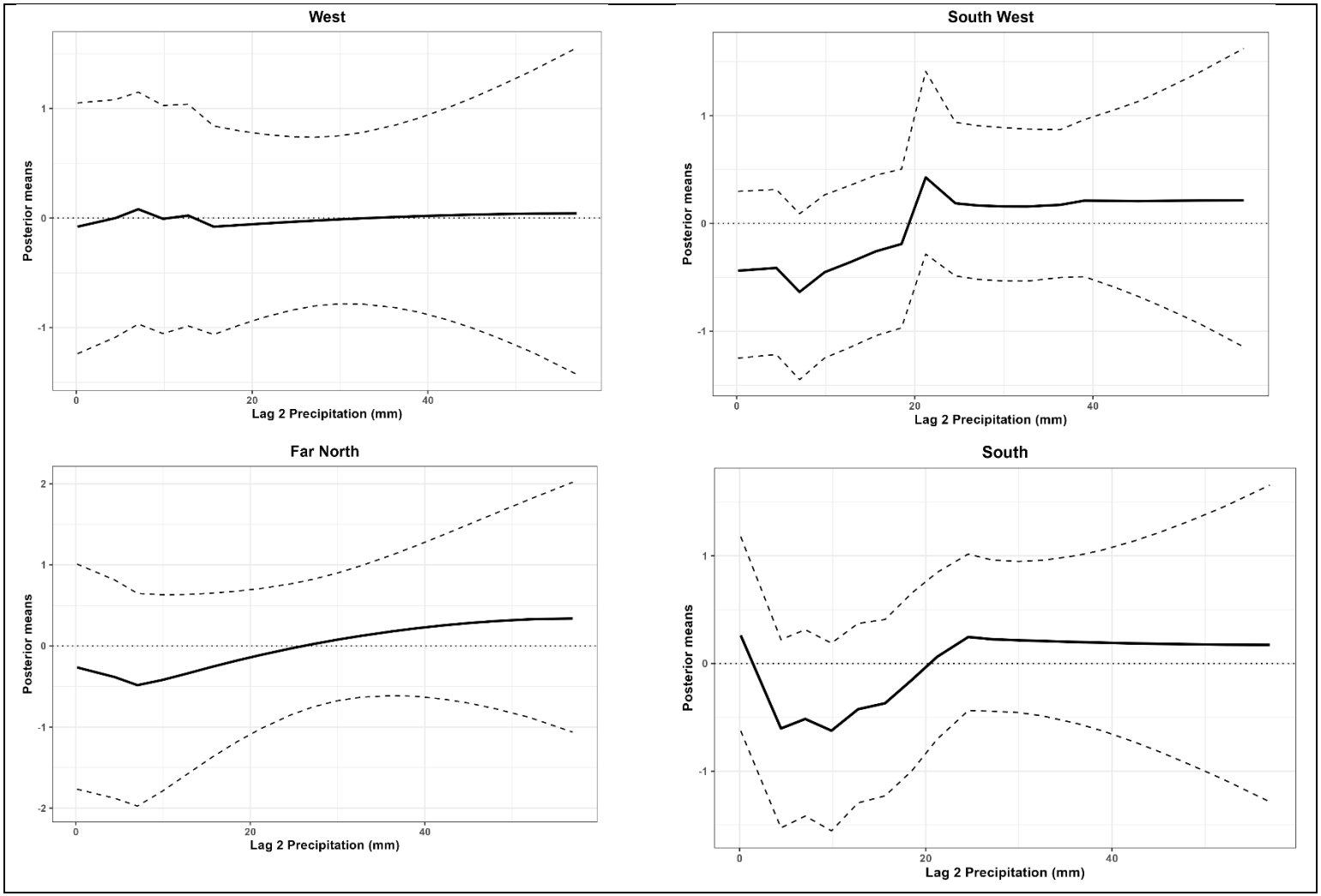
The non-linear effect of precipitation and temperature on cholera incidence in each administrative region was also inferred. Table 10 presents the estimated nonlinear effects of lagged precipitation on cholera risk across the administrative regions of Cameroon.

The Centre, East, and West regions demonstrated relatively stable associations across the precipitation range, suggesting limited association of cholera transmission with changes in precipitation. In contrast, nonlinear patterns were observed in the Littoral, North, South West, Far North, and South regions. In the Littoral region, cholera risk increased slightly at moderate precipitation levels before declining at approximately 35 to 40 mm and increasing again at higher precipitation levels.

The North region showed elevated risk under low precipitation conditions, followed by a decline as precipitation increased. The South West and South regions exhibited reduced risk at lower precipitation levels and increased association at moderate precipitation levels before stabilising at higher precipitation amounts. In the Far North region, relative risk increased gradually with increasing precipitation, suggesting a stronger sensitivity of cholera transmission to rainfall compared with the other regions.

Table 10: Nonlinear effects of precipitation on cholera incidence across regions in Cameroon. Solid lines show posterior mean (β) and dashed lines show the 95% credible intervals derived from Model 3. The horizontal dotted line indicates no effect at $\beta = 0$.



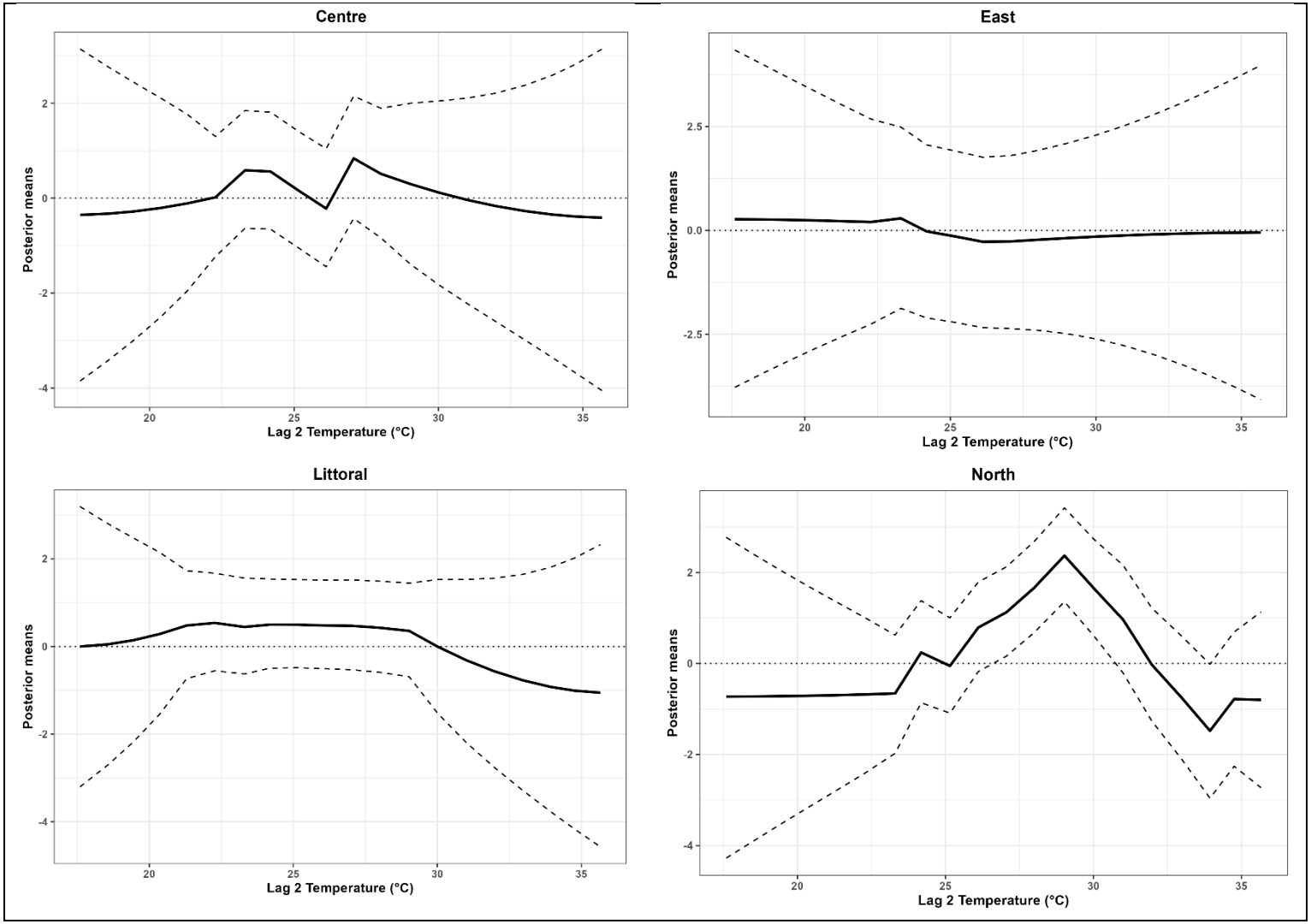

West
Posterior means
Lag 2 Precipitation (mm)
South West
Posterior means
Lag 2 Precipitation (mm)
Far North
Posterior means
Lag 2 Precipitation (mm)
South
Posterior means
Lag 2 Precipitation (mm)

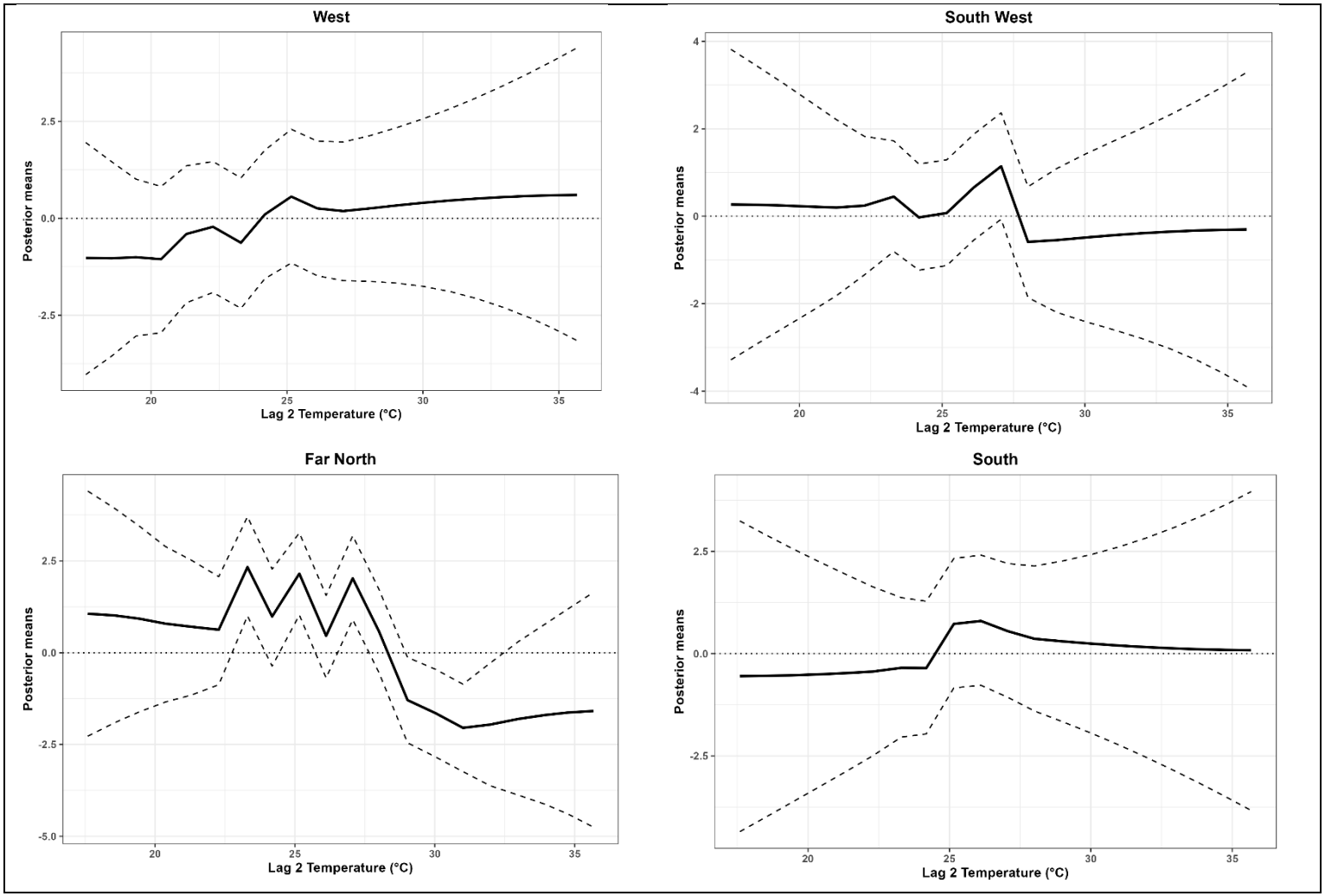
The nonlinear effects of lagged temperature was observed in Table 11. In several regions, including the Centre, East, Littoral, West, South West, and South, the posterior mean effects remained close to zero across most of the temperature range, with wide credible intervals including zero. This indicates limited evidence of a nonlinear association between temperature and cholera incidence in these regions.

More pronounced temperature effects were observed in the northern regions. In the North region, a major increase in cholera risk between approximately $25°C$ and $30°C$ was observed, reaching its highest posterior effect near $29°C$ before declining at higher temperatures. Similarly, the Far North exhibited elevated posterior effects at temperatures between approximately $23°C$ and $28°C$, after which the effect decreased as temperature increased from $29°C$. Although the credible intervals remained wide for some temperature ranges, these patterns suggest that moderately high temperatures may create environmental conditions favourable for cholera transmission in the northern regions.

The observed decline in posterior effects at the highest temperatures may indicate that excessively hot conditions become less favourable for the environmental persistence of *V.cholerae* or may reflect changes in human behaviour, water availability, or reporting patterns during extreme heat conditions. Overall, the temperature analyses suggest that temperature influences on cholera transmission are more evident in the northern regions of Cameroon, whereas little evidence of a strong temperature effect was observed in the southern and coastal regions.

Table 11: Nonlinear effects of temperature on cholera incidence across regions in Cameroon. Solid lines show posterior mean (β) and dashed lines show the 95% credible intervals derived from Model 3. The horizontal dotted line indicates no effect at $\beta = 0$.




West
Posterior means
Lag 2 Temperature (°C)
South West
Posterior means
Lag 2 Temperature (°C)
Far North
Posterior means
Lag 2 Temperature (°C)
South
Posterior means
Lag 2 Temperature (°C)

4.3.5. Temporal random effects

Figure 8 shows the regional structured temporal effects (RW1) on cholera risk across Cameroon between 2018 and 2023. Positive temporal effects indicate periods when cholera incidence was higher than expected after accounting for covariate effects and spatial dependence, whereas negative values indicate lower than expected incidence.

Temporal patterns varied across regions. The Centre, Littoral, and South West regions had recurrent periods of elevated temporal risk throughout the study period. In the South West and Littoral regions, positive temporal effects became more pronounced after 2020, while the Centre region showed alternating periods of increased and reduced temporal risk. The South and West regions also experienced elevated temporal effects, particularly during the later years of the study period.

In contrast, the northern regions demonstrated weaker and less persistent temporal effects. The Far North showed brief periods of elevated temporal risk in 2018 to 2019 and 2023, whereas the North region changed from positive temporal effects in 2018 - 2019 to mostly negative effects after 2020. The East region remained relatively stable throughout most of the study period, with temporal effects remaining close to zero.

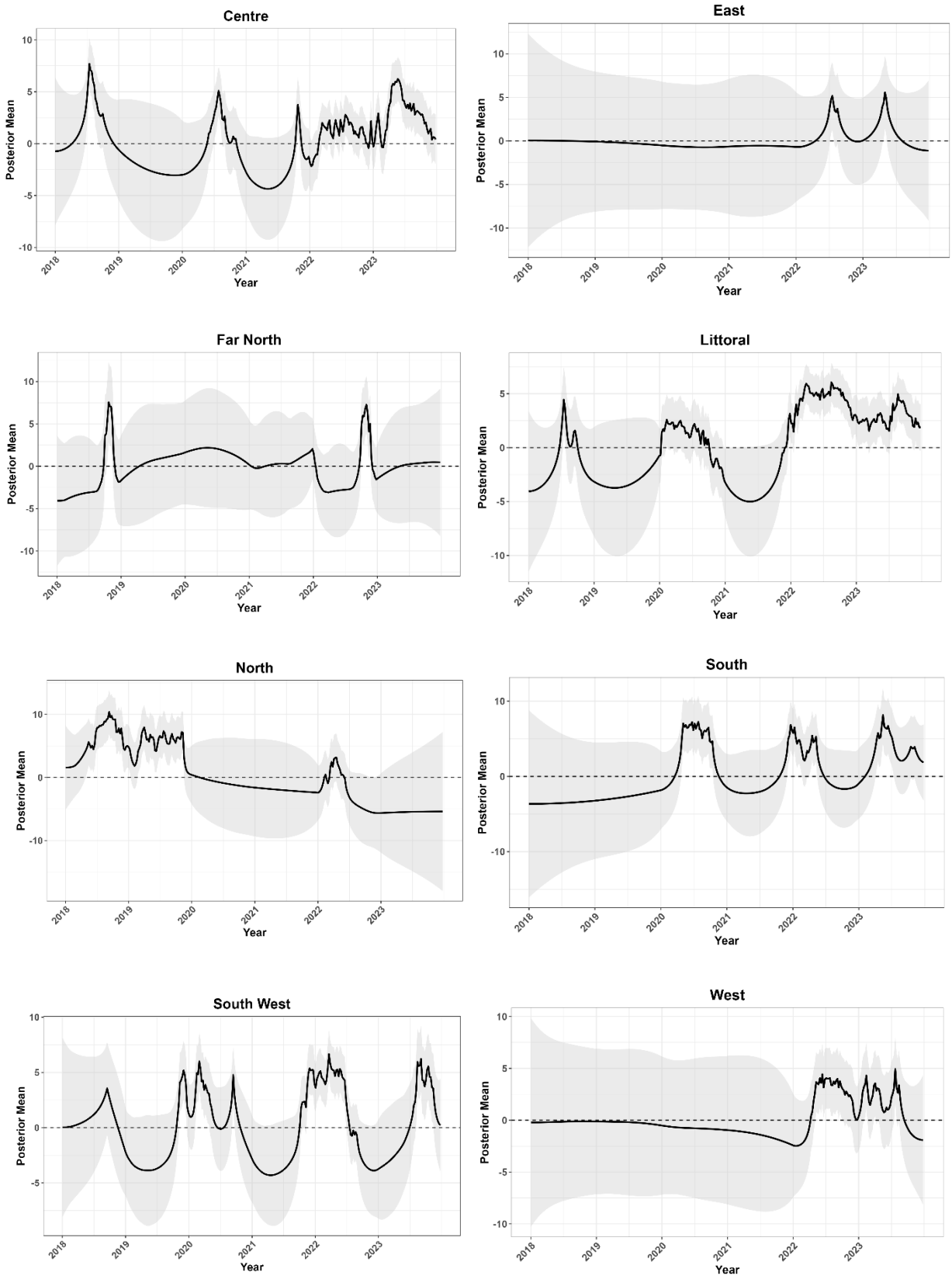


Figure 8: Region-specific posterior structured temporal effects (RW1) with 95% credible intervals from Model 3, 2018–2023.

4.4. Objective 3: To map the distribution of cholera relative risk across health districts health districts in Cameroon

To further investigate the spatial structure of cholera transmission, posterior estimates of the relative risks associated with the spatially structured random effects and corresponding hotspot probabilities were examined (Figure 9 and Figure 10). These maps represent residual spatial variation in cholera risk after accounting for the effects of the included covariates, temporal trends, and space-time interaction.

4.4.1. Relative risk maps

The residual relative risk maps derived from the Model 2 spatially structured random effects and Model 4 were compared visually (Figure 9). In 2018, elevated residual risks were concentrated primarily in western and south western Cameroon, with additional high risk districts observed in the Far North region. A total of 25 districts exhibited relative risks greater than five. In 2019, elevated residual risk persisted in both the northern and southern parts of the country, with more pronounced clusters occurring in the northern districts and several districts within the South West region.

During 2020 and 2021, the elevated residual risks became concentrated in the southern part of Cameroon, particularly within the West, South West, Littoral, and Centre regions. These clusters contained numerous neighbouring districts with relative risks exceeding two and, in several cases, five. In contrast, much of the Far North, North, and Adamawa regions continued to exhibit relatively low residual risk. By 2022 and 2023, the spatial pattern became more pronounced and fragmented. Although western and south western Cameroon remained important areas of elevated residual risk, several isolated districts with high relative risks also emerged in the Far North, Centre, and southern regions. Overall, the maps indicate persistent spatial clustering of cholera risk, with elevated residual risk occurring predominantly within the western and southern districts of Cameroon throughout the study period.

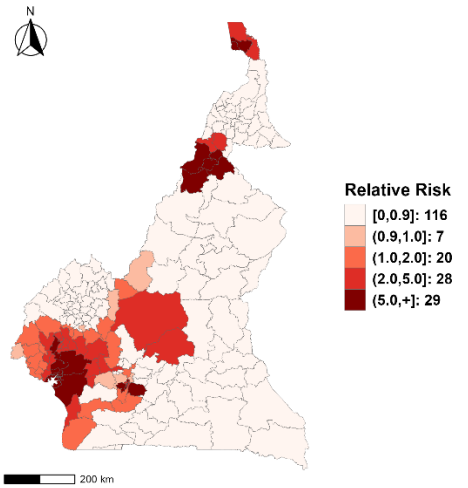
Comparing the spatial random effect maps from Model 2 (Maps on left side of Figure 9) and Model 3 (Maps on right side of Figure 9), showed differences in both the magnitude and distribution of residual cholera risk. After adding the covariates mentioned in $\eta_{ijktg} = \alpha_g + f(\text{Prec}_{kt}) + f(\text{Temp}_{kt}) + \beta_1(\text{Slope}) + \beta_2(\text{PSW}) + \beta_3(\text{Poor sanitation}) + \beta_4(\text{Poverty index}) + \beta_5(\text{Hcf}) + U_{jg} + V_{jg} + \gamma_{tk} + \phi_{tk}$ Equation 28, several districts showed reduced relative risks compared with Model 2. This was particularly evident in 2018, 2019, and 2020, where the number of districts with relative risks greater than five decreased after covariates were added. For example, in 2020 the number of districts with relative risks exceeding five declined from 53 in Model 2 to 33 in Model 3, indicating that part of the elevated spatial risk observed in these districts was explained by the included covariates.

Conversely, some districts exhibited higher residual relative risks in Model 3 than in Model 2. These increases occurred where the included covariates are predicting lower risk than expected, resulting in a greater proportion of the remaining variation being captured by the spatial random effect. As a result, some districts moved from lower risk categories in Model 2 to moderate or higher categories in Model 3.

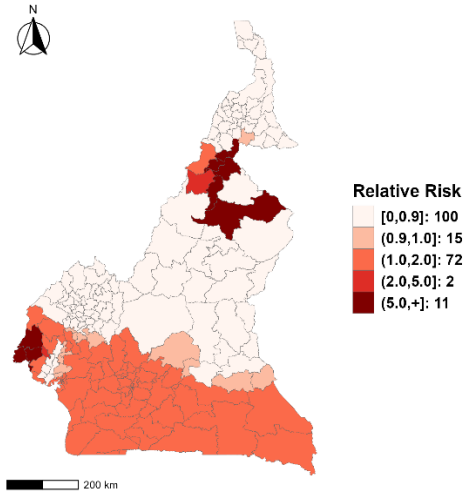
We can determine that the inclusion of covariates helped explain part of the spatial variation in cholera risk that was previously captured by the random effects alone. This was confirmed by the reduction in DIC and the decrease in the proportion of variance captured by the spatially structured component, which declined from 0.999 in Model 2 to 0.819 in Model 3. However, some residual spatial variation remained unexplained by the covariates added.

Left: Model 2 structured spatial random effects

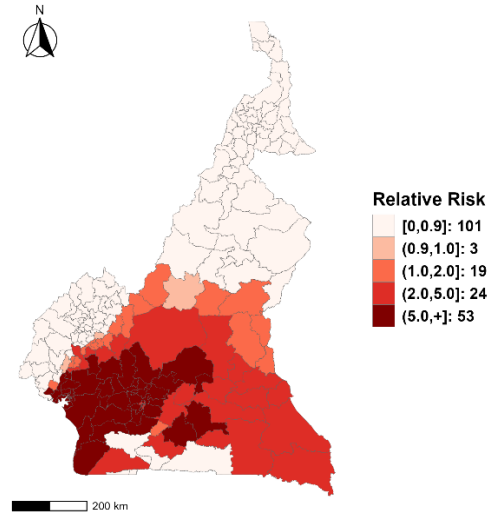
Relative Risk Associated with Spatial Random Effects 2018



Relative Risk Associated with Spatial Random Effects 2019

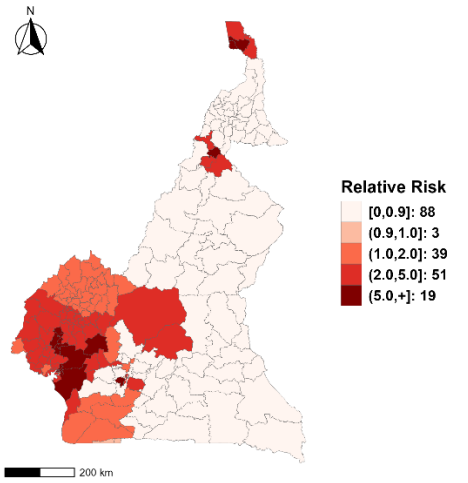


Relative Risk Associated with Spatial Random Effects 2020

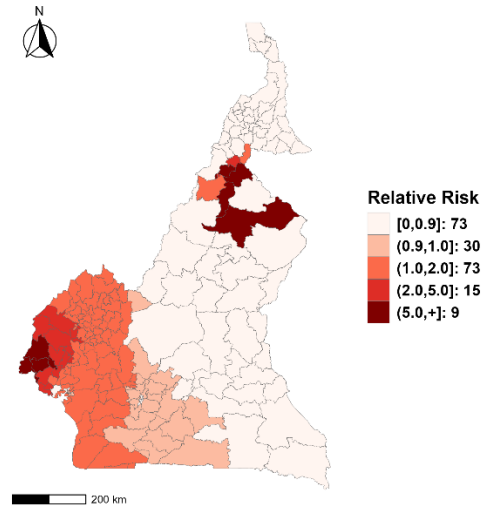


Right: Model 3 structured spatial random effects

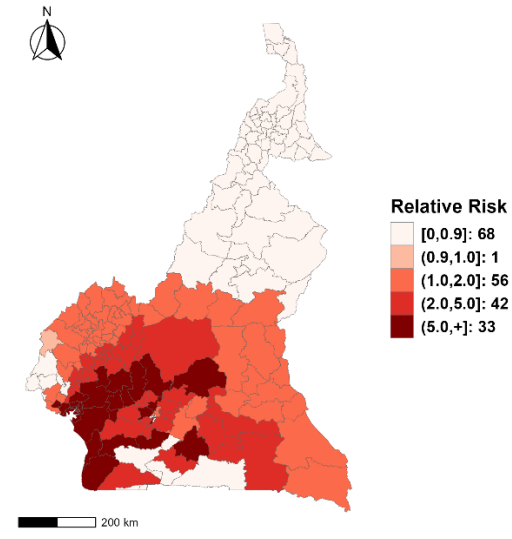
Relative Risk Associated with Spatial Random Effects 2018



Relative Risk Associated with Spatial Random Effects 2019



Relative Risk Associated with Spatial Random Effects 2020



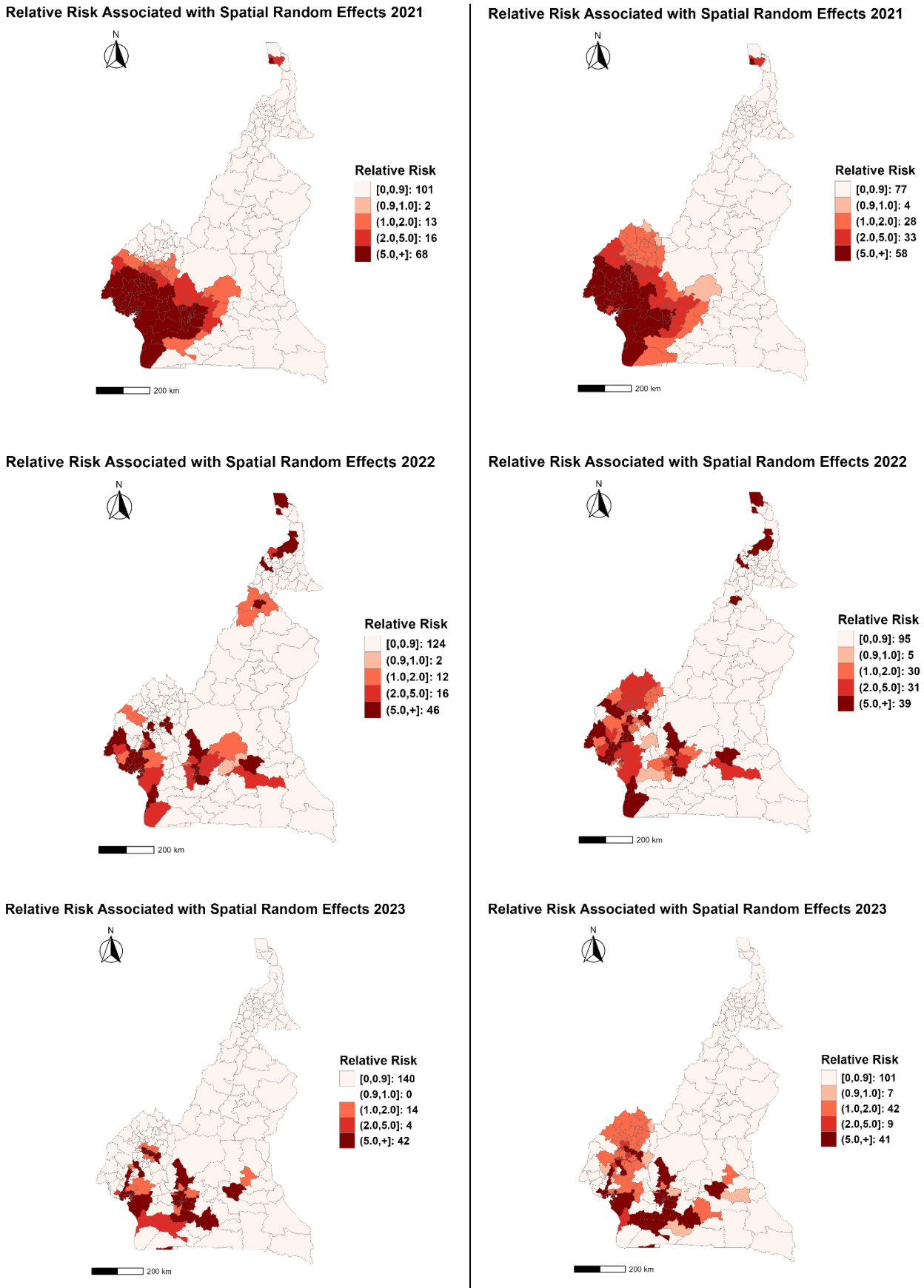


Figure 9: Spatially structured relative risk maps inferred from Model 2 (left) and Model 3 (right).

4.4.2. Exceedance probability

The posterior exceedance probability maps derived from Model 3 estimates, showed the spatial distribution of districts with strong evidence of higher residual cholera risk, expressed as the probability that the district-specific relative risk exceeded one (Figure 10). In 2018, districts with high exceedance probabilities ($P(RR > 1) > 0.8$) were concentrated primarily in western and south western Cameroon, with a smaller cluster occurring in the Centre and North regions. By 2019, the pattern shifted towards the northern part of the country, where several districts in the Far North and North regions showed high exceedance probabilities, while a smaller hotspot showed persistence in the South West region.

During 2020 and 2021, the strongest evidence of elevated residual risk was observed in districts in the South West, Littoral, South, and Centre regions. Large clusters of neighbouring districts exhibited exceedance probabilities greater than 0.8, indicating a high degree of certainty that cholera risk remained above the national average after accounting for the included covariates and temporal effects. In contrast, much of northern Cameroon displayed low exceedance probabilities during these years, suggesting limited evidence of elevated residual risk.

In 2022 and 2023, although western and south western Cameroon continued to show many districts with high exceedance probabilities, additional hotspots emerged in the Far North, Centre, and southern regions. Overall, the exceedance probability maps complemented the relative risk maps by demonstrating persistent spatial clustering of elevated residual cholera risk throughout the study period.

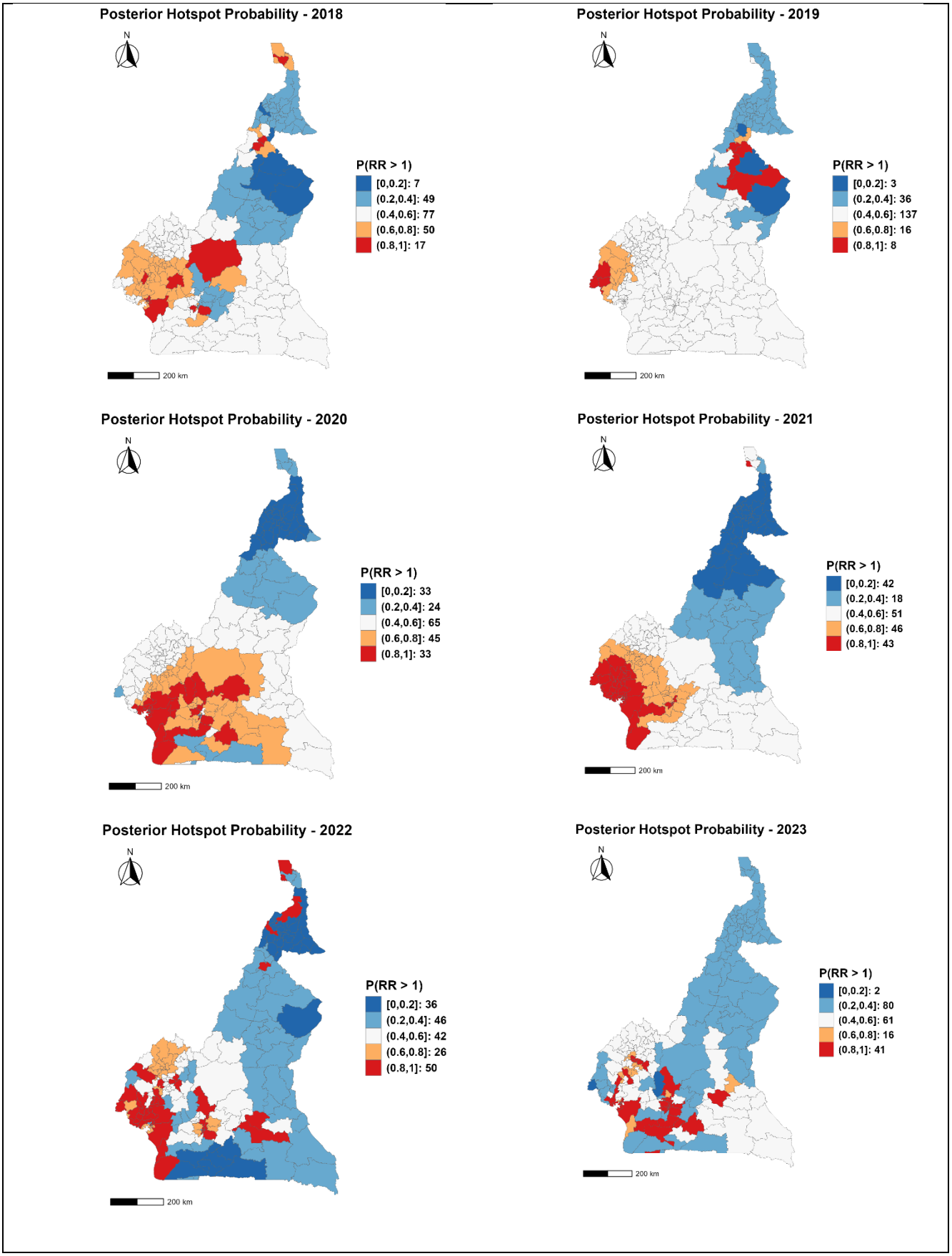


Figure 10: Posterior exceedance probability maps inferred from Model 3 showing the probability that district-level relative risk exceeds one, $P(RR > 1 \mid y)$, across health districts in Cameroon from 2018 to 2023.

5. DISCUSSION

Cholera spatio-temporal risk distribution was explored in this study, with goal to contribute knowledge to fighting the ongoing cholera global public health challenge .This study investigated the spatio-temporal dynamics of cholera risk in Cameroon between 2018-2023, while accounting for interactions among environmental, hydroclimatic, and socioeconomic factors. Several Bayesian spatio-temporal models were evaluated to assess how different model structures and explanatory variables influenced observed cholera risk patterns and trends. This chapter critically reflects on the findings in relation to the research objectives and discusses the strengths, limitations, and practical implications of the study.

5.1. Covariate selection and multicollinearity

Modelling cholera risk requires an understanding of the epidemiological processes that drive disease transmission. Consequently, the study began with an exploratory assessment of eleven predictor variables selected based on their established associations with cholera risk in the literature. Among these variables, building density and poverty index were identified as highly correlated with population density, with building density exceeding the acceptable variance inflation factor threshold. Areas experiencing socioeconomic development often undergo rapid urban expansion, resulting in increased population and building densities (Santos-Vega et al., 2016). The removal of highly collinear variables improved model stability and reduced redundancy among predictors.

Spearman correlation analysis revealed generally weak associations between individual covariates and cholera incidence. This finding is consistent with the multifactorial nature of cholera transmission, whereby disease occurrence is shaped by the interaction of multiple environmental, climatic, socioeconomic, and behavioural drivers rather than by a single dominant factor (Elimian et al., 2020). For example, districts with relatively high wealth indices may still contain densely populated informal settlements characterised by poor sanitation and limited access to safe water. Such complex interactions are difficult to capture using simple bivariate relationships. Furthermore, several risk factors may exhibit nonlinear effects or delayed influences on cholera transmission, making their contributions less apparent during exploratory analyses (Alcayna et al., 2022; Girotto et al., 2024). These findings suggested adopting modelling approaches capable of accounting for both these nonlinear relationships and spatial and temporal dependencies.

5.2. Research objective 1: Evaluating the temporal autoregressive process governing cholera transmission across time

To explore delayed effects and temporal persistence in cholera transmission, an autoregressive temporal process was incorporated into the modelling framework. The results demonstrated strong temporal dependence, whereby cholera incidence in a given week was strongly influenced by incidence in the preceding week and, in turn, influenced transmission in subsequent weeks. Similar findings were reported by Shackleton et al. (2023), who identified an AR(1) process when investigating climatic influences on cholera transmission in Kolkata. This pattern was observed across most regions reporting cholera cases, with the exception of the East Region, where comparatively few cases were recorded throughout the study period.

The strong temporal persistence observed across most regions is consistent with the epidemiological characteristics of *V.cholerae*. Infected individuals may excrete the bacterium in highly infectious quantities

for up to ten days, while the incubation period typically ranges from 12 hours to five days (Montero et al., 2023). These characteristics facilitate rapid environmental contamination and sustained transmission, such that periods of elevated cholera incidence are likely to be followed by continued transmission in subsequent weeks. Delays in case detection and reporting within Cameroon's passive surveillance system may further contribute to this persistence by postponing the implementation of public health interventions, thereby allowing transmission to continue before control measures are initiated.

Although autoregressive models provide valuable insight into outbreak persistence and short term forecasting, the remaining unexplained variation highlights the need to consider broader spatial and environmental processes. These findings therefore justify the development of the district level Bayesian spatio-temporal disease mapping framework presented under Research Objective 2, which incorporates spatial dependence and additional risk factors to better explain cholera transmission patterns across Cameroon health districts.

5.3. Research objective 3: Assessing the effects of selected hydroclimatic, environmental, and socio-economic covariates on the spatio-temporal distribution of cholera incidence

5.3.1. Model performance

The spatio-temporal modelling process began with an examination of the Standardised Morbidity Ratio (SMR) to explore the observed distribution of cholera risk across Cameroon. The SMR maps provided an initial indication of spatial variation in disease burden by comparing observed and expected cholera cases across health districts. The results demonstrated that elevated cholera risk was not confined to a single region but shifted spatially throughout the study period. In particular, 2022 and 2023 exhibited the greatest number of districts with SMR values exceeding two, indicating widespread elevated risk during these years.

To further investigate these patterns, Bayesian spatio-temporal disease mapping models were developed using cholera surveillance data and a range of environmental, hydroclimatic, socioeconomic, and healthcare-related covariates. Three model specifications were evaluated to determine the relative contribution of fixed effects and random effects in explaining cholera risk variation. The final model, which incorporated explanatory variables alongside spatial and temporal random effects, provided the best fit to the observed data, as indicated by the lowest DIC. Compared with models containing only random effects, the improved fit suggests that the selected covariates contributed meaningful information for explaining the spatio-temporal variation in cholera risk across Cameroon.

These findings are consistent with previous Bayesian disease mapping studies. For example, Li et al. (2025) compared several Bayesian spatio-temporal models for brucellosis risk in China and similarly found that models incorporating spatio-temporal interactions achieved superior model fit compared with simpler alternatives. The improved performance of the full model in the present study therefore supports the importance of jointly accounting for environmental risk factors, spatial dependence, and temporal variation when modelling infectious disease transmission.

The Poisson modelling framework performed adequately despite the presence of numerous zero counts within the surveillance data. The inclusion of spatially structured and unstructured random effects allowed the model to account for unobserved heterogeneity and potential overdispersion, resulting in good agreement between observed and predicted cholera incidence. Although the large number of zero observations may suggest the potential suitability of a zero-inflated modelling approach, such models require strong epidemiological justification for distinguishing between structural zeros and sampling zeros (Blangiardo & Cameletti, 2015; Osei et al., 2022). Structural zeros occur in locations where disease

transmission is not possible, whereas sampling zeros occur in susceptible locations where no cases happen to be observed during a particular period. In the context of Cameroon, health districts reporting zero cholera cases cannot be considered free from transmission risk. Historical outbreaks have demonstrated that cholera can emerge in districts that previously reported no cases, indicating that observed zeros are more likely to reflect temporary absence of transmission rather than structural immunity to disease occurrence (Mboringong et al., 2024).

5.3.2. Fixed effects

The inclusion of environmental, socioeconomic, and healthcare-related covariates improved model performance and provided insight into the factors associated with cholera risk across Cameroon. While the direction and magnitude of associations varied among variables, the results highlight the complex and context-dependent nature of cholera transmission.

Among the fixed effects, slope exhibited a significant negative association with cholera risk. Districts characterised by steeper terrain generally experienced lower cholera risk, whereas low-lying areas showed elevated risk. This finding is consistent with the role of topography in influencing drainage patterns and water accumulation. Areas with gentle slopes are more prone to surface water stagnation and flooding, which can facilitate contamination of water sources and increase exposure to *V.cholerae*. Similar relationships between topography and cholera transmission have been reported in Zimbabwe, where low elevation and poor drainage conditions were associated with increased disease risk (Luque Fernandez et al., 2012). Furthermore, low-lying areas often support larger human settlements and higher population densities, creating favourable conditions for disease transmission (Kruger et al., 2022).

Contrary to expectations, poverty and unimproved sanitation were negatively associated with cholera risk. This finding appears inconsistent with the established understanding that poor communities generally experience greater exposure to unsafe water, inadequate sanitation, and limited hygiene infrastructure (Asantewaa et al., 2024). However, this apparent contradiction may reflect the spatial scale of the analysis and the unique characteristics of urban cholera transmission in Cameroon. Although wealthier districts may have higher average socioeconomic indicators, they often contain densely populated informal settlements characterised by overcrowding, inadequate waste disposal, intermittent water supply, and limited sanitation services. Such conditions create environments highly favourable for cholera transmission despite higher average district-level wealth. Consequently, the negative association observed for poverty and sanitation variables may reflect the concentration of reported outbreaks within rapidly urbanising districts rather than a genuinely protective effect of improved socioeconomic conditions.

Water surface proportion showed a positive association with cholera risk, although with greater uncertainty than some other predictors. This finding is biologically plausible, as aquatic environments represent natural reservoirs for *V.cholerae*. A larger proportion of surface water within a district may increase opportunities for environmental persistence of the pathogen and contamination of drinking water sources. Previous studies have similarly reported positive associations between reliance on surface water and cholera occurrence (Wen et al., 2025). The relationship may become particularly important following extreme rainfall events, which can transport contaminated faecal material into rivers, lakes, and other water bodies through runoff and flooding.

Although hospitalisation rate was not included in the final model specification, it consistently emerged as one of the strongest predictors during model development. The variable greatly improved model fit with the lowest DIC, but also reduced the visibility of nonlinear hydroclimatic effects, suggesting that it captured a large proportion of the variation in reported cholera incidence. This likely reflects the passive nature of cholera surveillance in Cameroon, whereby cases are only recorded when individuals seek

treatment at healthcare facilities. Consequently, hospitalisation rate may capture not only true disease burden but also variation in healthcare access, health-seeking behaviour, outbreak detection capacity, and reporting completeness. This interpretation is supported by evidence from the Far North Region, where Yakum et al. (2017) reported that many suspected cholera patients initially sought treatment from alternative sources before attending healthcare facilities, while long travel distances further delayed access to care. Such barriers can contribute to underreporting in facility-based surveillance systems. Nonetheless, hospitalisation rate is a valuable variable when using as a proxy for cholera transmission in hospitals. This is supported by the case in Cameroon where to lack of preparedness to cholera outbreaks due to unequipped WASH infrastructure (Ateudjieu et al., 2019b).

5.3.3. Nonlinear effects

Several regions, including Centre, Littoral, North, South, South West, and Far North, exhibited increased cholera risk at specific rainfall levels. In most cases, relative risk increased with rainfall up to a certain threshold, after which the relationship either stabilised or declined. This pattern suggests that rainfall can influence cholera transmission through multiple mechanisms operating simultaneously. Moderate rainfall may increase transmission by facilitating the transport of faecal contaminants into water sources through runoff, while excessive rainfall may dilute bacterial concentrations or disrupt environmental conditions favourable for pathogen persistence.

The nonlinear rainfall relationship was particularly evident in the Littoral Region, a highly urbanised coastal area characterised by humid conditions and belonging to the Equatorial Monsoon climatic zone. This region receives some of the highest annual rainfall totals in Cameroon, ranging between approximately 2,000 and 10,000 mm annually (Ngwa et al., 2016). The observed plateau and subsequent decline in relative risk at higher rainfall levels may reflect dilution effects, whereby large volumes of water reduce the concentration of *V.cholerae* in aquatic reservoirs and lower the probability of human exposure (Magers et al., 2025; Powers et al., 2023).

The rainfall patterns observed in this study are consistent with the seasonal dynamics of cholera transmission reported across Cameroon. The country spans multiple climatic zones, ranging from humid equatorial conditions in the south to semi-arid Sudano-Sahelian conditions in the north. Consequently, rainfall can influence cholera risk through different pathways depending on location and season. In southern regions, intense rainfall events may increase contamination of water sources through flooding and surface runoff. In contrast, prolonged dry periods may increase bacterial concentrations in rivers, lakes, and other water bodies by reducing water volume. These contrasting mechanisms may explain the nonlinear rainfall effects observed across regions and highlight the importance of considering seasonal and geographical context when evaluating climate-disease relationships.

Temperature effects showed mostly linear relationships across most regions. However, pronounced increases in relative risk at higher temperatures were observed in the North and Far North regions. These regions belong to the Sudano-Sahelian climatic and Tropical Humid zone respectively and experience higher temperatures and lower rainfall than southern Cameroon . Elevated temperatures are known to enhance the growth, survival, and environmental persistence of *V.cholerae*, potentially increasing opportunities for transmission through contaminated water sources. The observed temperature patterns are consistent with previous studies demonstrating nonlinear temperature effects on cholera transmission. For example, Shackleton et al. (2023) reported that cholera risk increases sharply once temperatures exceed approximately 29-30$^{\circ}C$, suggesting the existence of critical climatic thresholds beyond which environmental conditions become increasingly favourable for bacterial proliferation. Similar findings have been reported across several cholera-endemic regions, where warmer temperatures promote bacterial growth, increase pathogen survival in aquatic reservoirs, and accelerate transmission processes.

Importantly, the influence of rainfall and temperature should not be considered independently of seasonal conditions. Previous research has shown that rainfall often exerts stronger nonlinear effects during wet seasons, whereas temperature effects may become more pronounced during dry seasons (Shackleton et al., 2023). This seasonal distinction is particularly relevant in Cameroon, where transmission pathways differ between climatic zones. Ngwa et al. (2016) reported that northern regions frequently experience elevated cholera incidence during dry periods, while southern regions are more strongly influenced by intense rainfall events and flooding. The regional variability observed in this study therefore reinforces the importance of accounting for nonlinear and geographically varying climatic effects when modelling cholera risk.

5.3.4. Temporal random effect

The temporal random effects captured residual variation in cholera risk over time after accounting for the observed covariates and spatial structure. Temporal patterns differed across regions, indicating that cholera transmission dynamics varied throughout the study period and were influenced by factors beyond those included in the model. These may include in increased reporting after first few reported cases, intervention measures, healthcare-seeking behaviour, population movement, and other local scale socioeconomic drivers. Compared with the AR(1) model, the magnitude of the temporal random effect was generally reduced in the final Bayesian spatio-temporal model, suggesting that the inclusion of hydroclimatic covariates and spatial dependence explained part of the temporal variation previously captured by the latent temporal process. Nevertheless, the persistence of residual temporal fluctuations highlights the complexity of cholera transmission and indicates that additional factors not represented in the model continue to influence disease dynamics across Cameroon.

5.4. Research objective 3: Mapping the distribution of cholera relative risk across health districts health districts in Cameroon.

The spatial random effects (BYM) capture residual risk after accounting for covariates and temporal trends. The relative-risk maps found in Figure 9, consistently identify high-risk clusters within districts in the Southern and Northern regions of Cameroon, and more pronounced risk elevations in districts bordering the coast and coastal districts and districts towards the north bordering Nigeria or Chad. These findings reflect on the conflict in the North West and South West regions near Nigerian border, resulting in the displacement of people. Consequently, this led to overcrowding in the South West, Littoral and Centre regions which hosted internally displaced persons (Bangwen et al., 2024; Musa et al., 2022). Elevated risks in the Far North region during 2018 and 2019, distribution pattern consistent with the Sudano-Sahelian climate zone's exposure to seasonal flooding and its historically documented cholera burden (Djomassi et al., 2013).

This vulnerability, coupled with the climatic factors, poor WASH conditions, inequality in health care access, all favourable for cholera spread enabled the disease outbreak and spread (Musa et al., 2022). The exceedance probability maps supported these findings, by identifying strongest evidence of elevated risk (probability >0.8) found in some districts in the South-West, Litoral and northern regions. In 2022, only a smaller hotspot persisted in the South-West, and by 2023 no district reached very high exceedance probabilities. This temporal shift in risk cluster aligns with the overall drop in cases observed in the data. These spatial insights directly reflect the model outputs from the specified Bayesian framework and confirm that interventions should focus on the identified hotspots.

This study makes several important contributions to cholera risk assessment in Cameroon. It integrates temporal and spatial modelling within a Bayesian disease mapping framework at both regional and health

district scales. This approach has been rarely applied in the Cameroonian context where previous studies have largely been descriptive (Bangwen et al., 2024; Mboringong et al., 2024, 2025; Musa et al., 2022). By incorporating spatial random effects, temporal processes, fixed effects, and nonlinear hydroclimatic effects, the modelling framework accounts for unmeasured heterogeneity across both space and time. Second, the use of weekly surveillance data over a six-year period, combined with climatic, environmental, socioeconomic, and healthcare-related covariates, provides a detailed understanding of cholera risk dynamics.

5.5. Study Limitations

Despite the strong performance of the Bayesian spatio-temporal framework, several limitations should be acknowledged. First, the analysis relied on routine cholera surveillance data, which are likely affected by underreporting due to delayed healthcare seeking, use of alternative treatment sources, limited healthcare accessibility, and conflict-related barriers to reporting. Consequently, the reported incidence may underestimate the true cholera burden, particularly in remote and conflict-affected districts (Hierink et al., 2022; Yakum et al., 2017). As the study was based on reported cases, any spatial or temporal variation in reporting completeness may also influence the observed risk patterns. Moreover, the aggregation of data at the health district level may mask important local-scale variations. Large urban districts often contain both rich neighbourhoods and highly vulnerable informal settlements, meaning that district-level socioeconomic indicators may not adequately capture the conditions driving cholera transmission at community level. This limitation may partly explain some of the unexpected associations observed for poverty and sanitation variables.

Furthermore, although evidence suggests that population displacement and sociopolitical instability have influenced cholera transmission in Cameroon, particularly in the North West and South West regions, conflict intensity and population mobility data were not explicitly included in the model. Incorporating these factors could improve understanding of the mechanisms underlying some of the observed spatial risk patterns.

Finally, while the model demonstrated good overall predictive performance, prediction accuracy declined in regions experiencing large outbreaks, particularly in highly urbanised areas. This suggests that epidemic spread and trends are influenced by additional processes not fully represented in the current framework, including behavioural responses, intervention measures, local transmission dynamics, and short-term changes in environmental conditions. Future studies should therefore explore the integration of higher-resolution socioeconomic data, conflict and mobility indicators, and additional environmental predictors to further improve cholera risk modelling and prediction in Cameroon.

5.6. Relevant insights for anticipatory action

Our results have several actionable implications. The consistently high risk in coastal Littoral, South-West and South suggests these areas should remain priorities for cholera prevention and preparedness. Resources for safe water, sanitation infrastructure, and oral cholera vaccine campaigns should be concentrated there, in line with WHO guidance that vaccines be coupled with WASH improvements in high-risk areas. The strong temporal dependence implies that rising case counts are a reliable early warning of continued transmission; surveillance systems should closely monitor trends week-to-week. The relatively small role of lagged climate in some regions means that early-warning systems in Cameroon must not rely on climate alone. Instead, they should integrate recent case data with local factors. Finally, our exceedance probability maps (uncertainty-weighted risk) can be directly used by the Cameroon Red Cross and health authorities to trigger interventions in districts that exceed risk thresholds with high confidence.

The Bayesian risk predictions generated by this study can directly inform cholera early-warning and anticipatory action systems. Anticipatory action involves the actions put in place to prevent or mitigate potential impacts of a hazard (Alcayna et al., 2025). The district-level posterior risk estimates and exceedance probabilities produced in this study are well suited to this purpose, as they communicate not only where risk is elevated but how certain that the elevation of the risk is. Environmental models that rely solely on climatic variables have accomplished real-time outbreak prediction but fail to capture transmission dynamics within affected populations, highlighting the need for frameworks that integrate both trigger and transmission components (Jutla et al., 2021). The BYM + RW1 framework developed here addresses this directly. Prediction of conditions favourable for cholera increased risk and identification of risk spatial clusters can allow timely allocation of resources and hence support the development of alert tools and mitigation strategies (Bordalo et al., 2021).

6. RECOMMENDATIONS FOR FUTURE RESEARCH

Future studies should explore the inclusion of conflict intensity, population displacement, and human mobility indicators to better understand their contribution to cholera transmission patterns in Cameroon. The elevated risk observed in districts affected by conflict and migration suggests that these factors may play an important role in shaping disease dynamics but were not explicitly represented in the present modelling framework.

Furthermore, future studies should examine interactions between seasonal conditions and hydroclimatic variables, as well as additional environmental predictors such as flooding, drought, soil moisture, and land use characteristics. Such analyses may provide a more comprehensive understanding of the nonlinear pathways through which climate and environmental conditions influence cholera transmission across Cameroon's diverse climatic zones.

Further methodological improvements could be achieved by exploring alternative Bayesian spatio-temporal model structures and interaction specifications. While this study employed a BYM spatial framework with a region-grouped temporal process, different forms of space-time interactions may capture disease dynamics more effectively by allowing spatial risk patterns to evolve differently across time. Future studies could compare alternative interaction structures, temporal processes, and hierarchical model specifications to evaluate their influence on model fit, predictive performance, and interpretation of cholera risk.

7. CONCLUSION

This study investigated the spatio-temporal dynamics of cholera risk in Cameroon between 2018 and 2023 using Bayesian spatio-temporal disease mapping models. The findings demonstrated that cholera transmission is shaped by a combination of temporal persistence, spatial clustering, environmental conditions, and contextual socioeconomic factors. Strong temporal dependence highlighted the importance of ongoing transmission processes, while the inclusion of hydroclimatic covariates explained part of the observed temporal variation. At the district level, the best-performing model incorporated spatial and temporal random effects alongside environmental and socioeconomic covariates, confirming

that both observed and unobserved factors contribute to cholera risk distribution. The analysis identified persistent high-risk areas in the southern coastal regions and northern border districts, with spatial patterns evolving throughout the study period. Nonlinear relationships between rainfall, temperature, and cholera risk further emphasised the importance of accounting for regional climatic variability rather than assuming uniform environmental effects across Cameroon. Although some socioeconomic indicators showed unexpected associations, these findings highlighted the influence of urbanisation, spatial aggregation, and reporting processes on observed disease patterns.

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8. APPENDIX

8.1. Appendix 1.

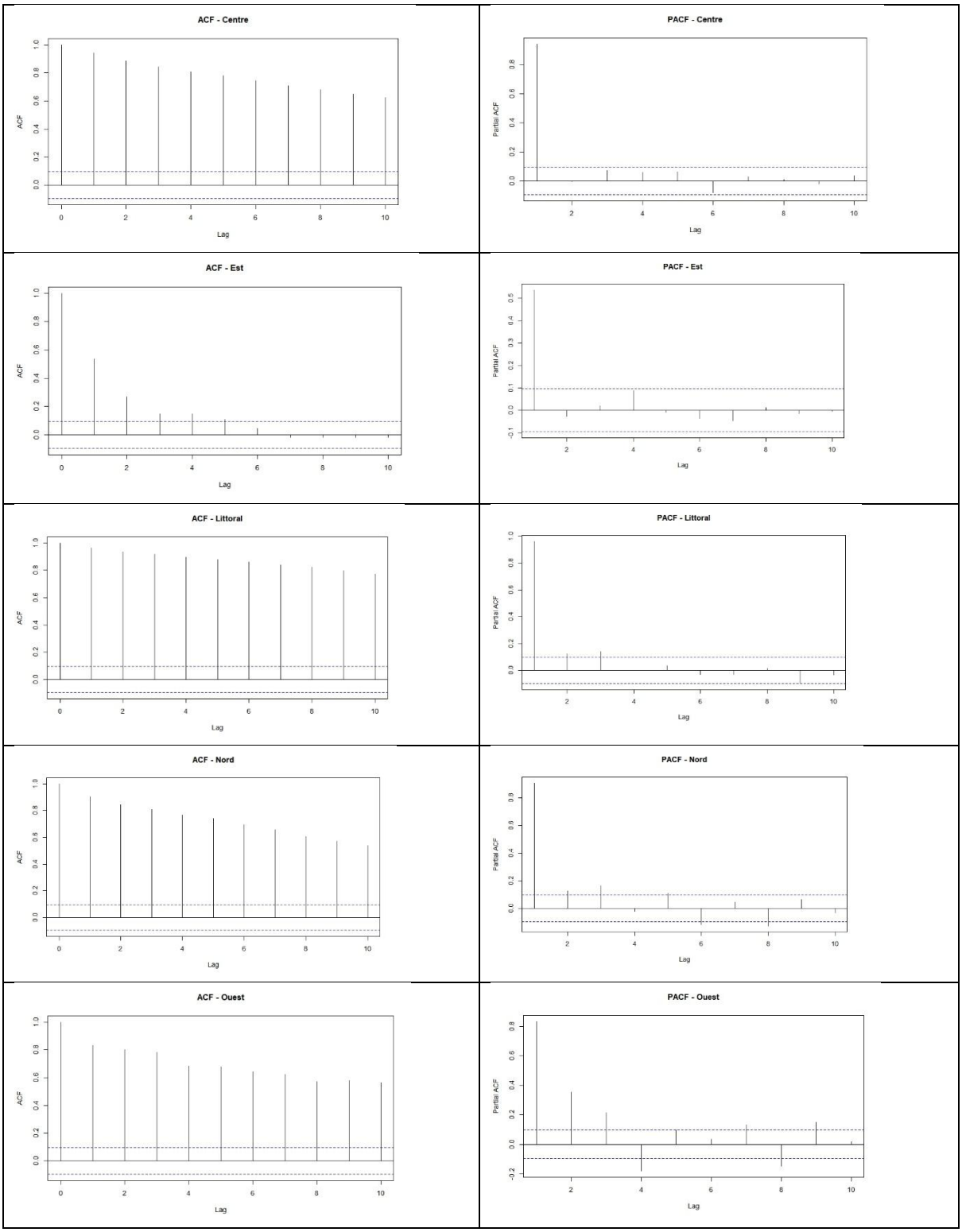


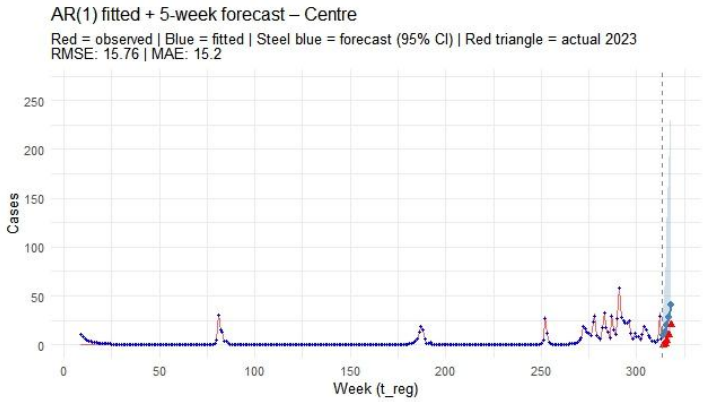
Figure 11: ACF (Left) and PACF (Right)

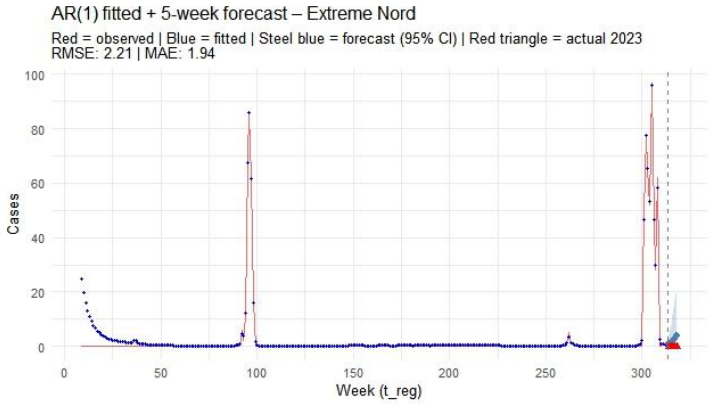
8.2. Appendix 2.

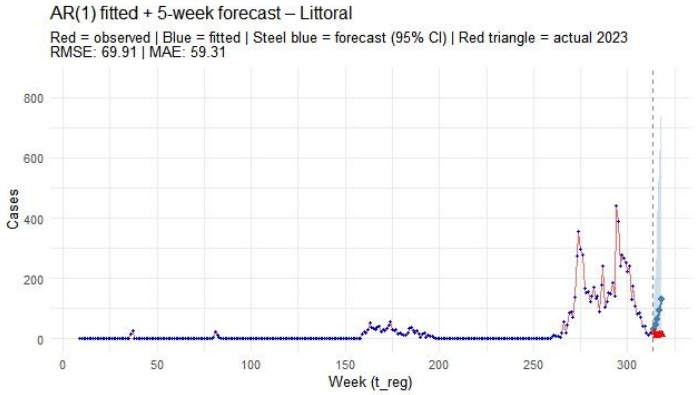
Table 12: Best-performing lagged hydroclimatic models for each administrative region based on the Deviance Information Criterion (DIC).

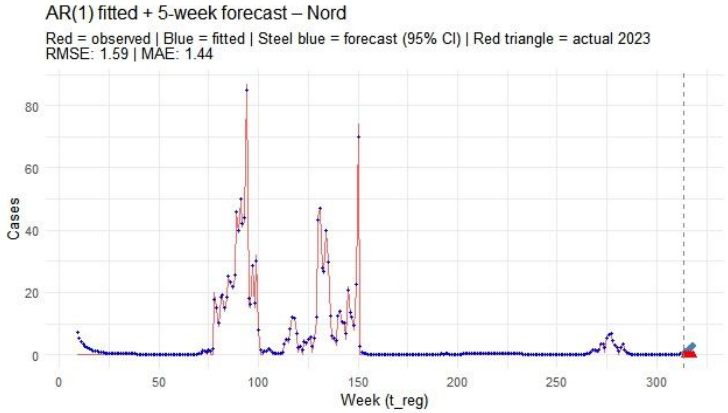
Region	Baseline AR1 DIC	Nonlinear Rain (Best Lag) DIC	Nonlinear Temperature (Best Lag) DIC
Centre	842.48	841.58 (lag1)	842.23 (lag2)
East	133.76	133.91 (lag1)	133.91 (lag2)
Littoral	1215.83	1217.98 (lag1)	1216.15 (lag1)
North	611.17	589.47 (lag1)	587.32 (lag2)
West	371.12	377.57 (lag1)	371.38 (lag2)
South West	670.3	665.22 (lag2)	665.01 (lag2)
Far North	587.49	569.64 (lag3)	572.50 (lag1)
South	506.79	503.86 (lag3)	509.46 (lag2)

8.3. Appendix 3.









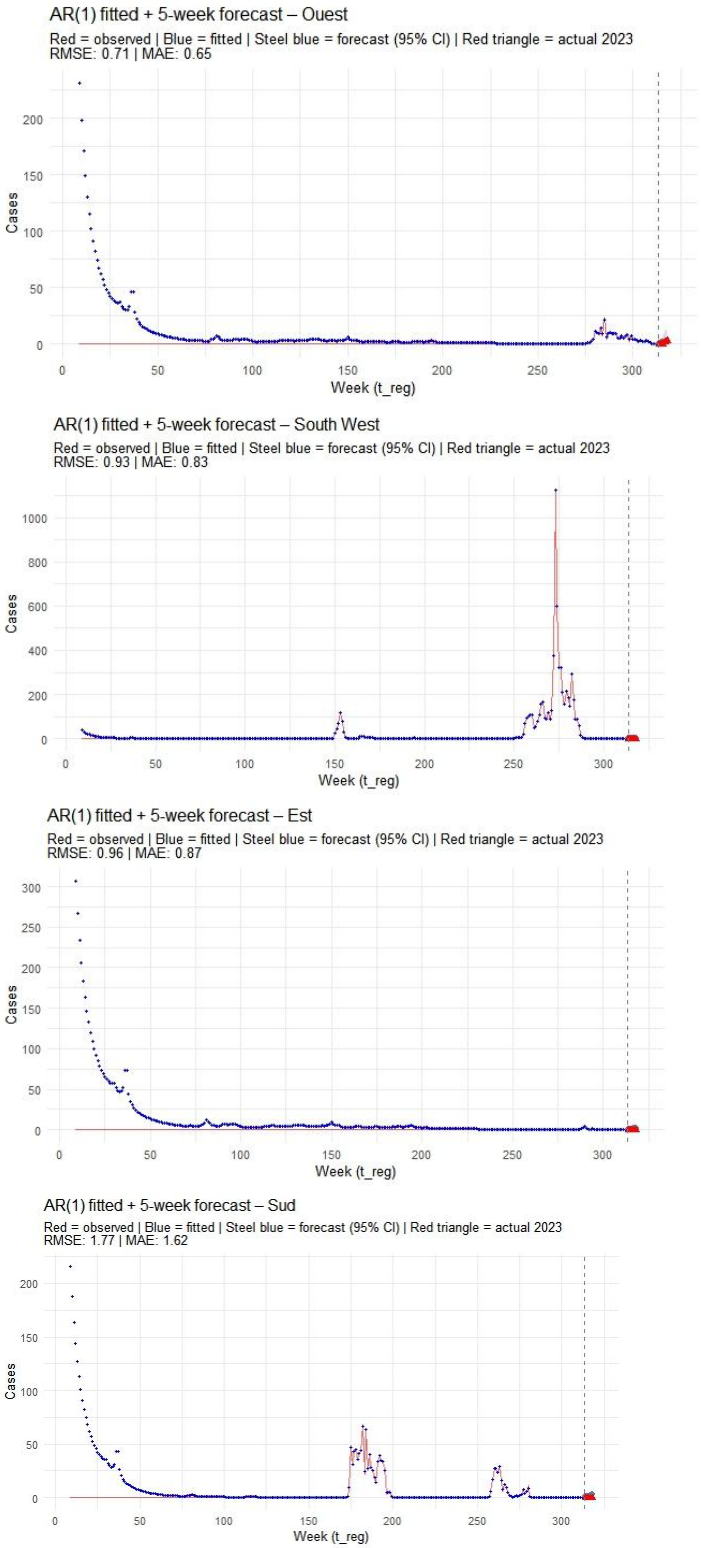
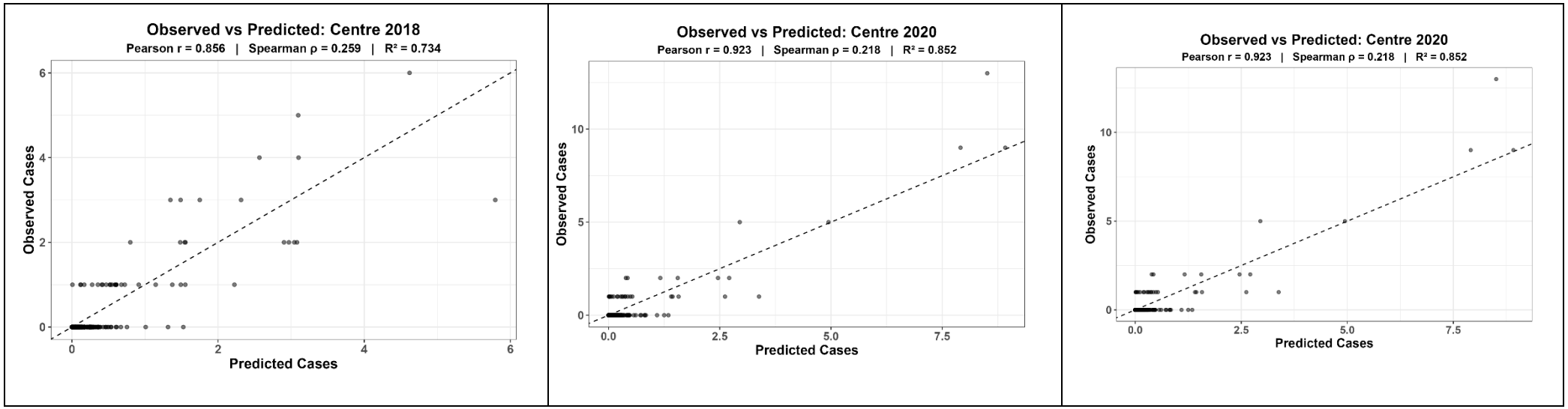
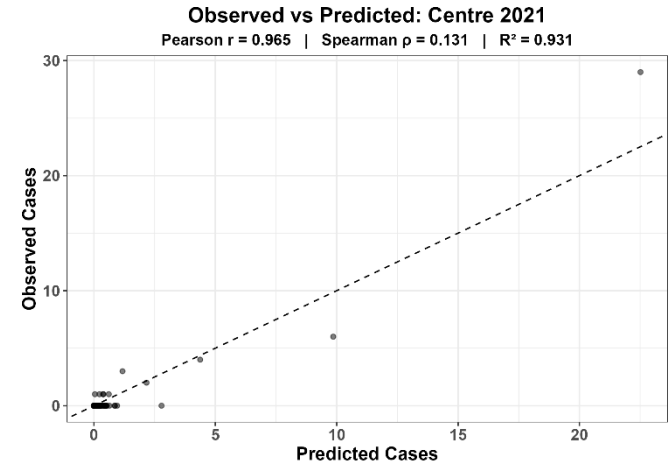


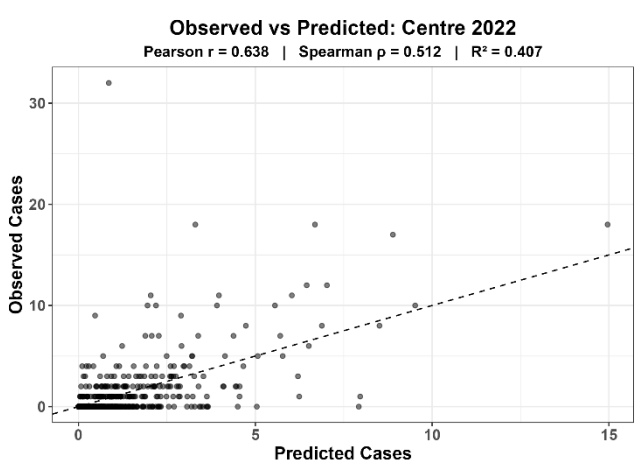
Figure 12: AR1 model fit (with 2018 to December 2022) and first 5 weeks forecast into 2023

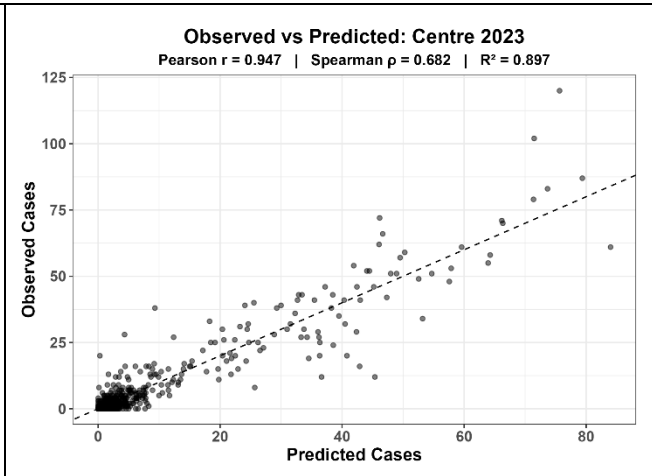
8.4. Appendix 4

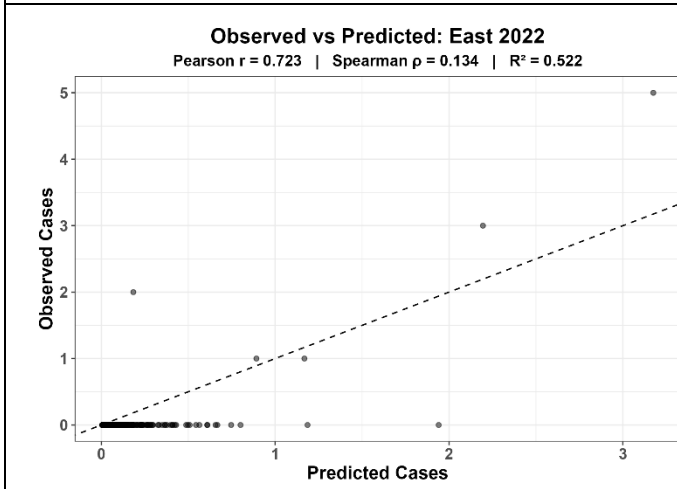
Table 13: Model 3 posterior mean estimates of model intercept and covariates for the spatio-temporal model

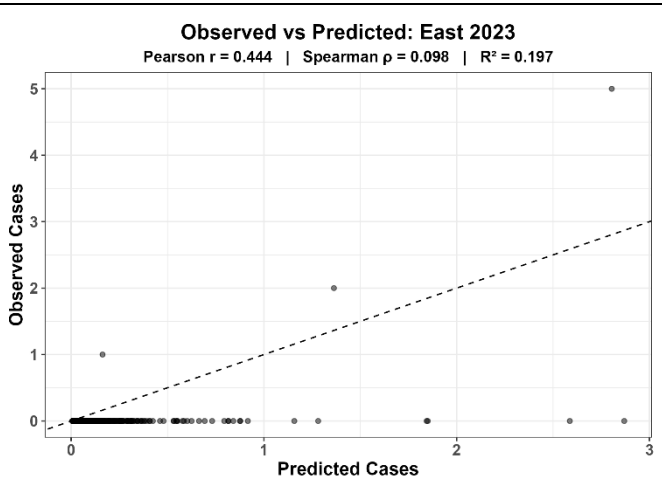


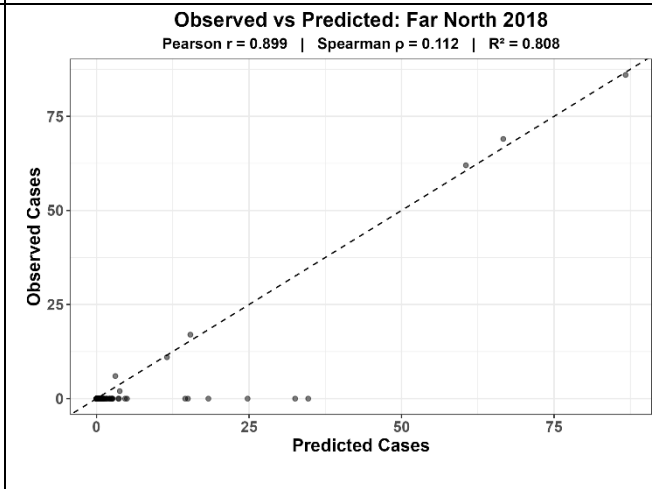

Observed vs Predicted: Centre 2021
Pearson r = 0.965 | Spearman ρ = 0.131 | R² = 0.931
Observed Cases
Predicted Cases

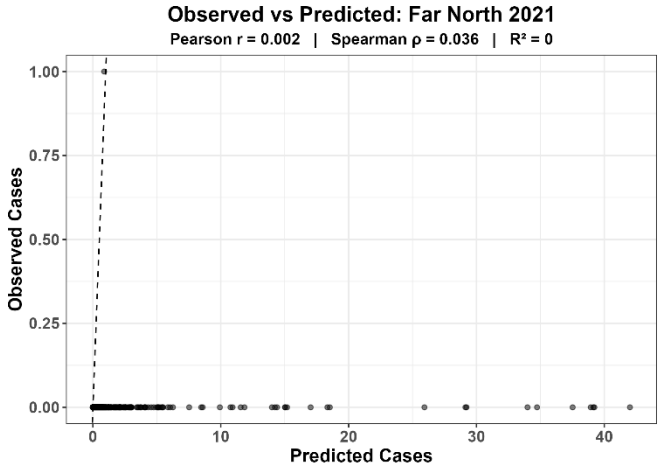

Observed vs Predicted: Centre 2022
Pearson r = 0.638 | Spearman ρ = 0.512 | R² = 0.407
Observed Cases
Predicted Cases

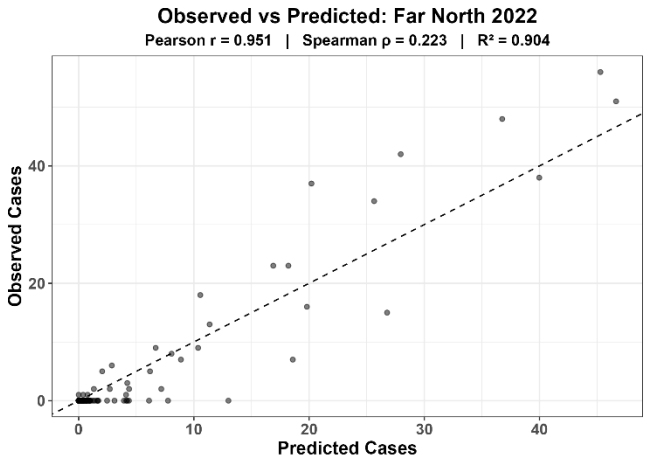

Observed vs Predicted: Centre 2023
Pearson r = 0.947 | Spearman ρ = 0.682 | R² = 0.897
Observed Cases
Predicted Cases

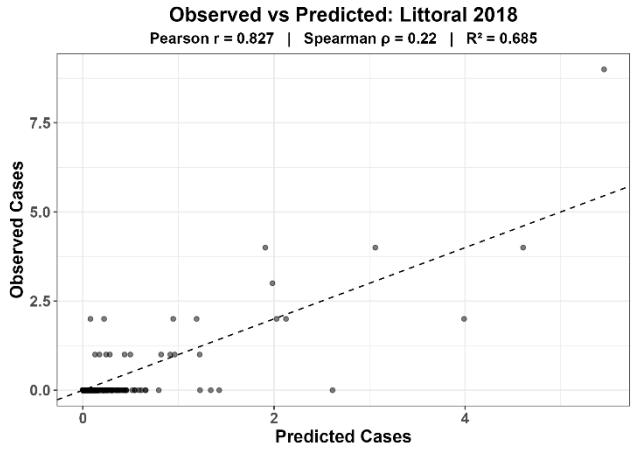

Observed vs Predicted: East 2022
Pearson r = 0.723 | Spearman ρ = 0.134 | R² = 0.522
Observed Cases
Predicted Cases

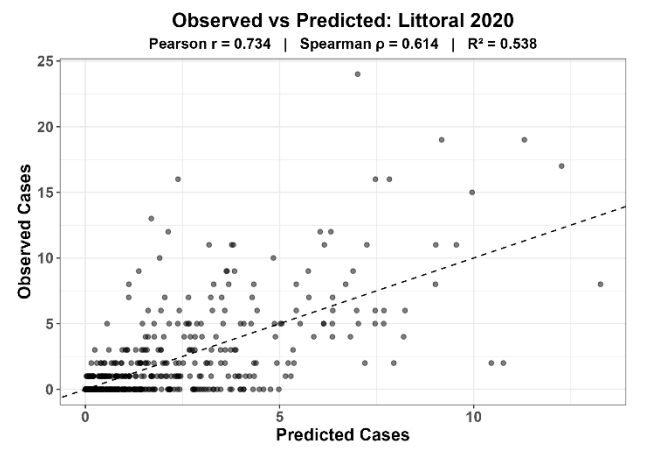

Observed vs Predicted: East 2023
Pearson r = 0.444 | Spearman ρ = 0.098 | R² = 0.197
Observed Cases
Predicted Cases

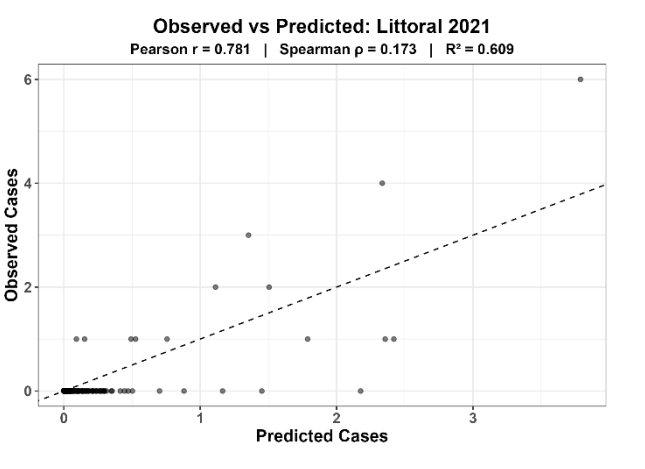

Observed vs Predicted: Far North 2018
Pearson r = 0.899 | Spearman ρ = 0.112 | R² = 0.808
Observed Cases
Predicted Cases

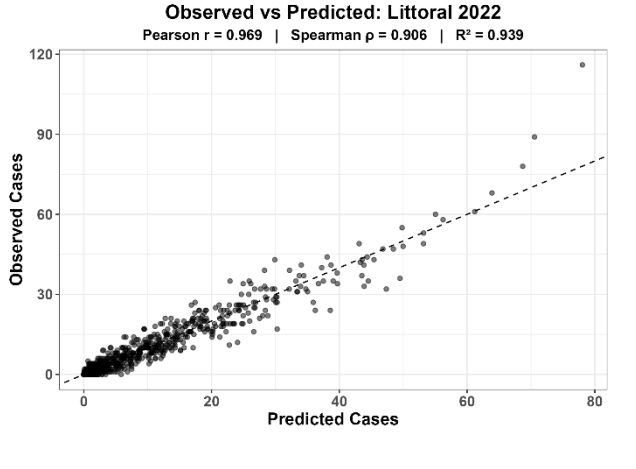

Observed vs Predicted: Far North 2021
Pearson r = 0.002 | Spearman ρ = 0.036 | R² = 0
Observed Cases
Predicted Cases

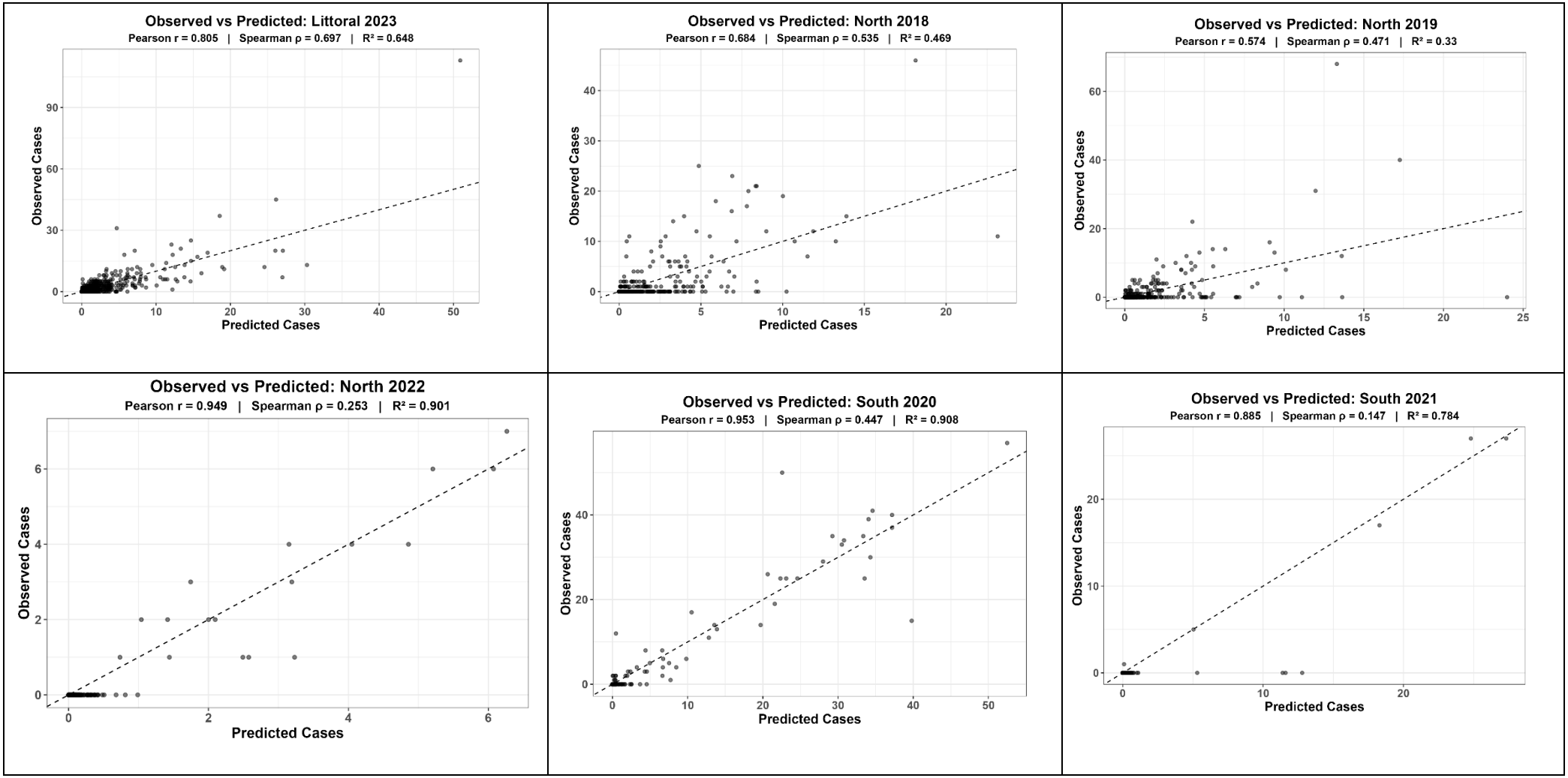

Observed vs Predicted: Far North 2022
Pearson r = 0.951 | Spearman ρ = 0.223 | R² = 0.904
Observed Cases
Predicted Cases

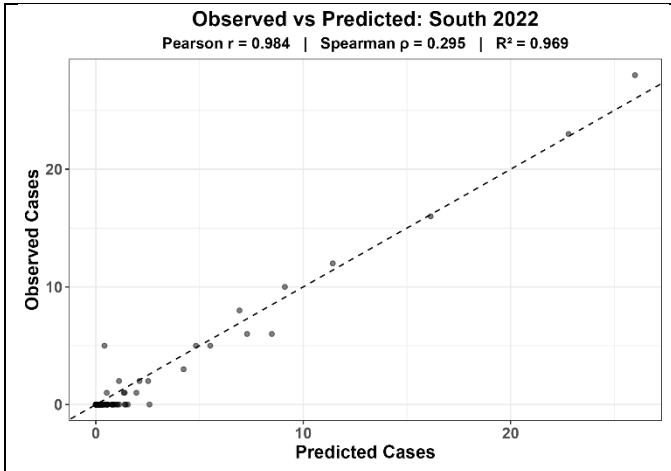

Observed vs Predicted: Littoral 2018
Pearson r = 0.827 | Spearman ρ = 0.22 | R² = 0.685
Observed Cases
Predicted Cases

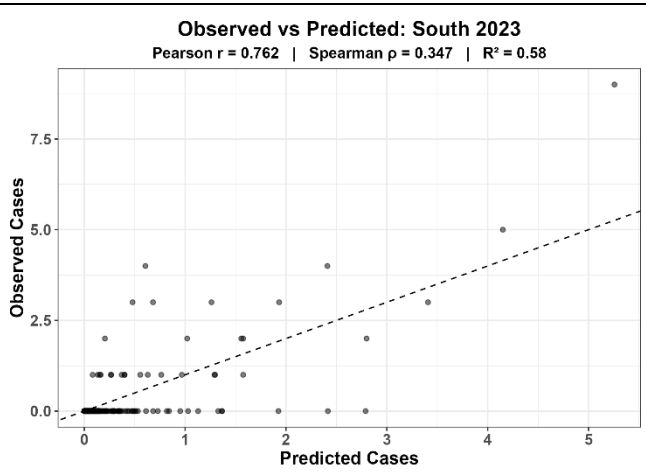

Observed vs Predicted: Littoral 2020
Pearson r = 0.734 | Spearman ρ = 0.614 | R² = 0.538
Observed Cases
Predicted Cases

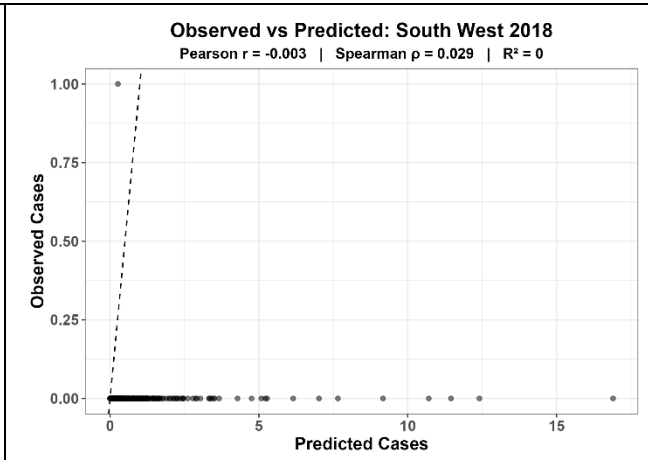

Observed vs Predicted: Littoral 2021
Pearson r = 0.781 | Spearman ρ = 0.173 | R² = 0.609
Observed Cases
Predicted Cases

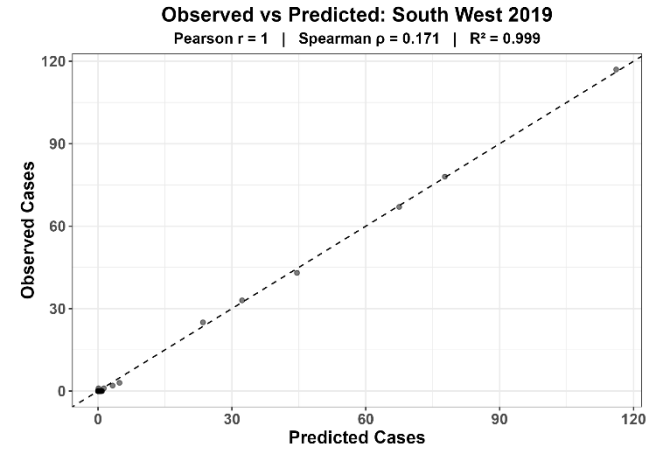

Observed vs Predicted: Littoral 2022
Pearson r = 0.969 | Spearman ρ = 0.906 | R² = 0.939
Observed Cases
Predicted Cases

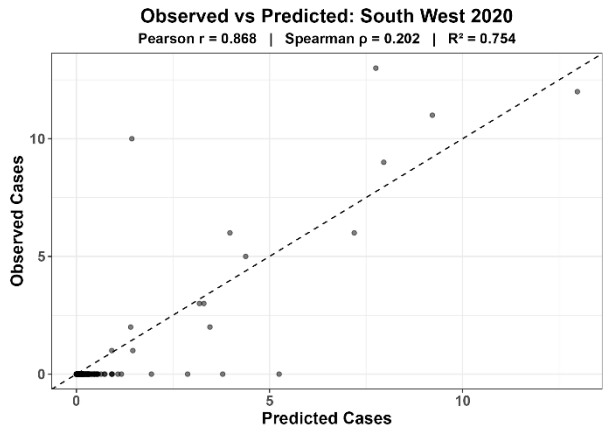

Observed vs Predicted: Littoral 2023
Pearson r = 0.805 | Spearman ρ = 0.697 | R² = 0.648
Observed Cases
Predicted Cases
Observed vs Predicted: North 2018
Pearson r = 0.684 | Spearman ρ = 0.535 | R² = 0.469
Observed Cases
Predicted Cases
Observed vs Predicted: North 2019
Pearson r = 0.574 | Spearman ρ = 0.471 | R² = 0.33
Observed Cases
Predicted Cases
Observed vs Predicted: North 2022
Pearson r = 0.949 | Spearman ρ = 0.253 | R² = 0.901
Observed Cases
Predicted Cases
Observed vs Predicted: South 2020
Pearson r = 0.953 | Spearman ρ = 0.447 | R² = 0.908
Observed Cases
Predicted Cases
Observed vs Predicted: South 2021
Pearson r = 0.885 | Spearman ρ = 0.147 | R² = 0.784
Observed Cases
Predicted Cases

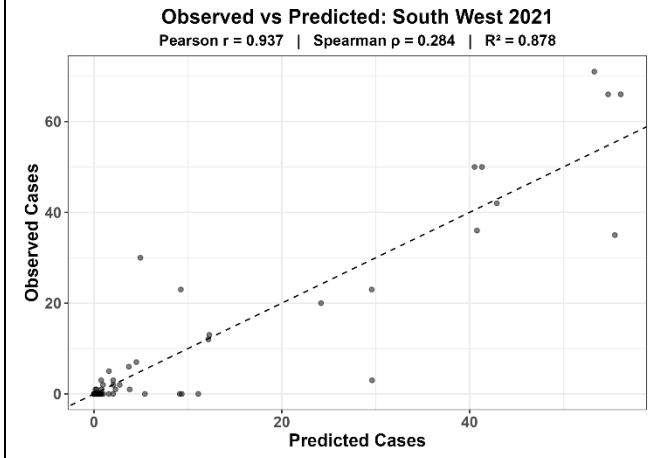

Observed vs Predicted: South 2022
Pearson r = 0.984 | Spearman ρ = 0.295 | R² = 0.969
Observed Cases
Predicted Cases
0
10
20

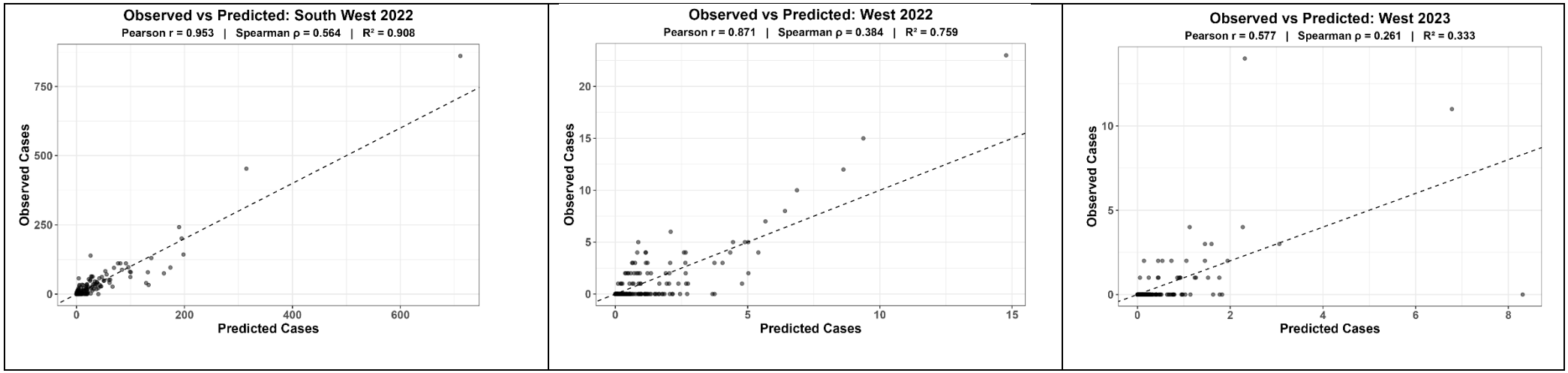

Observed vs Predicted: South 2023
Pearson r = 0.762 | Spearman ρ = 0.347 | R² = 0.58
Observed Cases
Predicted Cases
0.0
2.5
5.0
7.5
0
1
2
3
4
5


Observed vs Predicted: South West 2018
Pearson r = -0.003 | Spearman ρ = 0.029 | R² = 0
Observed Cases
Predicted Cases
0.00
0.25
0.50
0.75
1.00
0
5
10
15


Observed vs Predicted: South West 2019
Pearson r = 1 | Spearman ρ = 0.171 | R² = 0.999
Observed Cases
Predicted Cases
0
30
60
90
120


Observed vs Predicted: South West 2020
Pearson r = 0.868 | Spearman ρ = 0.202 | R² = 0.754
Observed Cases
Predicted Cases
0
5
10


Observed vs Predicted: South West 2021
Pearson r = 0.937 | Spearman ρ = 0.284 | R² = 0.878
Observed Cases
Predicted Cases
0
20
40
60


Observed vs Predicted: South West 2022
Pearson r = 0.953 | Spearman ρ = 0.564 | R² = 0.908
Observed Cases
Predicted Cases
0
250
500
750
0
200
400
600
Observed vs Predicted: West 2022
Pearson r = 0.871 | Spearman ρ = 0.384 | R² = 0.759
Observed Cases
Predicted Cases
0
5
10
15
20
0
5
10
15
Observed vs Predicted: West 2023
Pearson r = 0.577 | Spearman ρ = 0.261 | R² = 0.333
Observed Cases
Predicted Cases
0
5
10
0
2
4
6
8